



Degree Project in Financial Mathematics

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Modelling Credit Spread Risk in the Banking Book (CSRBB)

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Abstract

Risk measurement tools and strategies have until recently been calibrated for a low-for-long interest rate environment. However, in the current higher interest rate environment, banking supervisory entities have intensified their regulatory pressure on institutions to enhance their assessment and monitoring of interest rate risk and credit spread risk. The European Banking Authority (EBA) has released updated guidelines on the assessment and monitoring of Credit Spread Risk in the Banking Book (CSRBB), which will replace the current guidelines by 31st December 2023. The new guidelines identify the CSRBB as a separate risk category apart from Interest Rate Risk in the Banking Book (IRRBB), and specifies the inclusion of liabilities in the risk calculations. This paper proposes a CSRBB model that conforms to the updated EBA guidelines. The model uses a historical simulation Value at Risk (HSVaR) and Expected Shortfall (ES) approach, and includes a 90-day holding period, as suggested by Finansinspektionen (FI). To assess the effectiveness of the model, it is compared with a standardised model of FI, and subjected to backtesting. Additionally, the paper suggests modifications to the model to obtain more conservative results.

Keywords

CSRBB, Credit Spread Risk, Risk Modelling, EBA, Value at Risk, Expected Shortfall, Backtesting

Modellering av kreditspreadrisken i bankboken (CSRBB)

Sammanfattning

Riskmättningsverktyg och strategier har sedan nyligen anpassats till en lågräntemiljö. Dock till följd av den nuvarande högre räntemiljön har tillsynsmyndigheter för bankväsendet satt ökat tryck på institutioners utvärdering och rapportering av ränterisk och kreditspreadrisk. Den Europeiska Bankmyndigheten (EBA) har publicerat uppdaterade riktlinjer för bedömning och rapportering av kreditspreadrisken i bankboken (CSRBB), som ersätter de nuvarande riktlinjerna den 31 december 2023. De nya riktlinjerna identifierar CSRBB som en separat riskkategori från ränterisk i bankboken (IRRBB) och specificerar inkluderingen av skulder i riskberäkningarna. Denna uppsats föreslår en CSRBB-modell som följer EBAs uppdaterade riktlinjer. Modellen använder en Value at Risk (VaR) -metodik baserat på historiska simulationer och Expected Shortfall (ES), samt antar en 90-dagars innehavsperiod som föreslås av Finansinspektionen (FI). Modellens effektivitet utvärderas genom en jämförelse med FIs standardmodell för kreditspreadrisken i bankboken, samt genom backtesting. Slutligen diskuteras möjliga justeringar av modellen för att uppnå mer konservativa resultat.

Nyckelord

CSRBB, Kreditspreadrisk, Riskmodellering, EBA, Value at Risk, Expected Shortfall, Backtesting

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Acronyms

BCBS	Basel Committee on Banking Supervision
CS	Credit Spread
CSRBB	Credit Spread Risk in the Banking Book
EBA	European Banking Authority
ES	Expected Shortfall
EU	European Union
EVE	Economic Value of Equity
FED	Federal Reserve System
FI	Finansinspektionen
IRRBB	Interest Rate Risk in the Banking Book
Nibor	Norwegian Interbank Offered Rate
NII	Net Interest Income
NS	Nelson-Siegel
NSS	Nelson-Siegel-Svensson
Stibor	Stockholm Interbank Offered Rate
VaR	Value at Risk
ZCB	Zero Coupon Bond

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Chapter 1

Introduction

In this chapter, the background information necessary to understand the relevance of this thesis will be provided. The background section highlights the current high interest rate environment and credit spread levels, and introduces the regulatory environment for managing interest rate risk and credit spread risk. In the related work section, approaches to model credit risk and credit spread movements, are presented. Lastly, the chapter will present the purpose and contribution, objectives, and outline of this thesis.

The aim of this paper is to suggest a model for calculating an institution's exposure to CSRBB. The model includes several steps, such as the selection process of credit spread sensitive instruments on an institution's balance sheet, the construction of yield curves and calculation of credit spreads, and the implementation of a historical VaR (Value at Risk) method. The VaR calculation incorporates a 90-day holding period for the financial instruments. By applying this model, it becomes possible to forecast the CSRBB of a given portfolio with respect to both its value and profitability, providing a forward-looking projection for a 90-day period.

1.1 Background

Risk measurement tools and strategies have until recently been calibrated in a low-for-long interest rate environment. However, a combination of recent events, such as supply chain issues, the Covid-19 pandemic, and the Russian invasion of Ukraine, have led to interest rates and credit spreads reaching levels not seen in decades. This

is problematic in the current market environment, as high interest rates and credit spreads can severely impact the economic capital and earnings of an institution. As a result, central banks and supervisory authorities have made it a priority to enhance the management of these risk categories [7]. An event that illustrates the importance of risk management is the shutdown of Silicon Valley Bank in March 2023. In the previous low-interest rate environment, Silicon Valley Bank invested a large part of their customer deposits in long-maturity securities. However, when the United States Federal Reserve System (FED) increased interest rates, the value of the bank's securities decreased. In order to meet widespread customer withdrawals, the Silicon Valley Bank had to sell their securities and recognise great losses.

Credit spread risk has long been overlooked due to the misconception that changes in credit spreads have a price impact similar to parallel shifts in the risk-free curve [22]. As a result, it has generally been accepted to combine interest rate risk and credit spread risk. However, banking supervisory entities have intensified regulatory pressure on institutions, urging them to enhance their assessment and monitoring of both interest rate risk and credit spread risk. This pressure ultimately aims to increase the quality and resilience of financial structures within the industry.

The EBA is an independent agency of the European Union (EU) that works to ensure stability of the banking sector in the EU. The EBA is responsible for setting and monitoring the standards for banks and other financial service providers. Based on the framework EBA/GL/2018/02 (*“Guidelines on the management of interest rate risk arising from non-trading book activities”*)[2], EBA has in 2022 issued the updated guidelines EBA/GL/2022/14 related to the capital requirements of Article 84 of the EU directive 2013/36/EU (Capital Requirements Directives) on Interest Rate Risk (IRRBB) and Credit Spread Risk (CSRBB) arising from non-trading book activities. The 2018 publication of the EBA guidelines introduced regulations for banks to identify and manage CSRBB as a separate risk class, however without further specification on approach. The CSRBB segments in the updated guidelines of 2022 include some specifying criteria for the assessment and monitoring of CSRBB and will replace the 2018 guidelines by 31st December 2023.

1.2 Related Work

Risk measurement within the financial industry is widely explored and there exists several studies suggesting different approaches. Many studies have been done on Value at Risk (VaR) and Expected Shortfall (ES) risk models. Yasuhiro et al. (2002) performed a comparative analysis of VaR and ES under market stress [32]. By simulating asset returns and assuming a multivariate extreme-value distribution, they found that both VaR and ES may underestimate the risk of financial instruments with fat-tailed properties, although ES do capture the properties to a greater extent.

Andersson et al. (2000) developed a credit risk optimisation model with linear programming, where Monte Carlo simulations were used to generate the credit risk distribution. The authors state that traditional models assume the return-loss of the portfolio to be normally distributed, although credit losses are characterised by high probability of small earnings coupled with small probability of losing a large part of the investment [14]. This suggests that the distribution is highly skewed. In order to manage the skewness of the data, they measure the risk by performing Conditional Value at Risk (CVaR), also known as Expected Shortfall. Furthermore, the authors state that traditional VaR models have several limitations compared to CVAR models. Limitations in terms of optimisation are the VaR models' non-convex and non-smooth nature, as well as lack of sub-additivity. Additionally, VaR models do not provide information on how much the loss of an extreme event may be, in contrast to CVaR models.

Pownall and Huisman also discuss in their paper "*Measuring Credit Spread Risk: Incorporating the Tails*" (2002) [24] traditional quantitative credit risk models. They state that traditional models assume the changes in credit spreads to be normally distributed, which leads to the measurements of the risk to be grossly underestimated as it does not account for extreme events. In order to model the additional tail fatness, Pownall and Huisman used the statistical Extreme Value Theory in order to estimate the tail index, i.e. the degrees of freedom. The tail index is then used in order to parameterise the Student's t-distribution of data of daily changes in the credit spreads. They observed the estimation of spread risk being improved when the tail-fatness was incorporated.

Additionally, there have been several different ways proposed to model the movement of credit spreads. Romo (2014) [28], describes an approach based on the structural

model of credit risk modeling of Merton (1974) with a two-factor stochastic volatility specification. Within the Merton model framework it is possible to express the equity price as a call option on the firm's assets, where the asset value is assumed to follow a geometric Brownian motion. Romo states that empirical tests of traditional structural models of credit risk has been somewhat unsuccessful in modeling credit spreads since predicted spreads have been far below observed ones. Zhang et al. (2009) have previously introduced a single-factor stochastic volatility model with jumps within the structural model framework, which showed satisfactory results for higher credit rated companies, but not for companies with lower credit ratings [10]. However, Romo suggests a two-factor stochastic volatility model, where one factor determines the correlation between short-term assets return and variance, and the other factor determines the correlation between long-term asset returns and variance. Introducing two factors to describe the stochastic volatility in the asset return provides flexibility and makes it possible to generate a more flexible credit spread term structure to match the observed term structure. Two-factor specification allows for correlation between the asset return and the variance process.

Manzoni (2002) suggests modeling credit spreads with time-varying volatility models in order to capture the conditional variance nature of the credit spread generating process [23]. She states that credit spread levels and changes present a fat-tailed behaviour compared to a normal distribution. In order to arrive at the most appropriate time series model, statistical tests such as the Box-Jenkin test and Dickey-Fuller were applied to the data, where she tested for stationarity and autocorrelation. Her tests suggested evidence for nonlinear dependence, changing volatility, and high level of intertemporal dependence. Lastly, the data was tested for $ARCH(p)$ and $GARCH(p, q)$ models with different parameters.

As stated, there exists several studies focusing on credit risk modelling and how to model credit spread movements. However, to the best of our knowledge, there are no previous papers presenting a "beginning-to-end" credit spread risk model in accordance with the 2022 EBA guidelines. Specifically, no studies on the selection process of credit spread-sensitive instruments were found.

1.3 Thesis Purpose and Contribution

The aim of this thesis is to propose a model for the Credit Spread Risk in the Banking Book (CSRBB) that aligns with the 2022 guidelines issued by the European Banking Authority (EBA/GL/2022/14). Due to the ambiguity of the framework on CSRBB, this thesis will provide more clarity and understanding regarding these guidelines, and propose a method on how to handle credit spread risk in liability instruments on the balance sheet.

To conduct this research, certain limitations were established in terms of scope. Specifically, this study will only concentrate on the identification, modelling and computational aspects of CSRBB, while other regulatory sections such as management, governance and reporting will not be addressed.

1.4 Objectives

This thesis aims to provide the following:

- *Understanding of credit spread risk and its assessment in the banking sector;*
- *A model of Credit Spread Risk in the Banking Book (CSRBB) in accordance with the new EBA regulatory framework (EBA/GL/2022/14); and*
- *Factors that make the suggested model more conservative.*

1.5 Thesis Outline

The thesis is structured such that the reader will be provided with a more detailed description of the theoretical background in chapter 2. In chapter 3, the mathematical framework is presented, followed by a thorough explanation of how these concepts are applied in the final model in chapter 4. Chapter 4 also provides a step-by-step methodological approach, demonstrating the practical implementation of the model. Moving forward, chapter 5 presents the model results, alongside a comprehensive discussion offering valuable insights and analysis. Finally, chapter 6 concludes the thesis by providing conclusive remarks and key takeaways from the research.

Chapter 2

Theoretical Background

This chapter provides an in-depth exploration of the theoretical background of the degree project. The following sections will introduce the concepts of credit spread risk, including its elements and drivers. The chapter will also give a more complete background on the regulatory environment relating to CSRBB, including the main regulatory bodies, as well as specifics on current legal frameworks. Lastly, the risk measures required for the assessment of CSRBB are introduced.

2.1 Credit Spread Risk in the Banking Book

Credit Spread (CS) is defined as the difference in yield between a risky debt instrument, e.g. a corporate bond, and a benchmark debt instrument with the same maturity. The yield on a bond is the interest paid by the issuer of the bond to the bondholder, expressed as a percentage of the bond's face value. Credit spreads can be seen as the additional compensation demanded by the investor to compensate for holding either a more risky or less liquid asset [29]. For example, if a bond is more risky than a benchmark bond, the investor may require a higher return, or yield, in order to invest. This difference in yield between a risky bond and a risk-free bond is called credit spread and can be illustrated as in figure 2.1.1. Credit spreads thus measure the additional risk assumed by investors for holding a more risky investment.

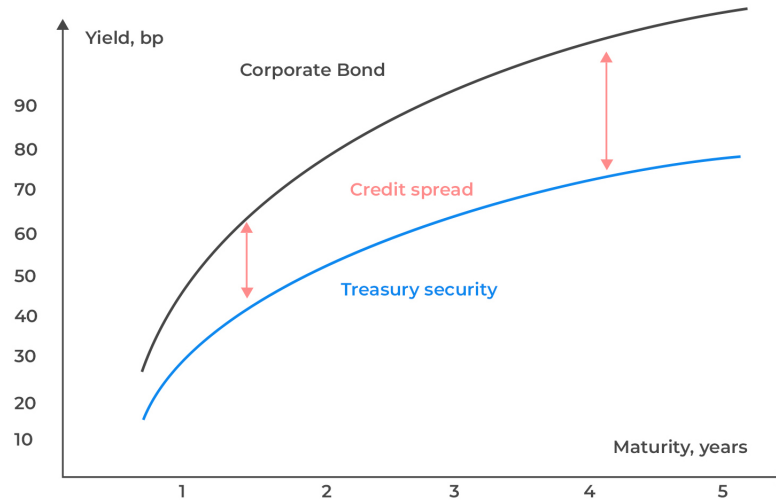


Figure 2.1.1: Credit Spread between a treasury and corporate bond [26].

The CSRBB refers to the sensitivity and exposure of the banking (or non-trading) book positions to CS widening and tightening. The banking book is a portfolio of financial instruments intended to be held until maturity, whereas the trading book positions refer to any securities not intended to be held until maturity, but rather traded regularly.

The credit spreads are said to widen when investors demand a higher yield, i.e. compensation, for holding the instrument, and tighten when investors are willing to accept a lower yield. The widening or tightening of credit spreads, which in this thesis will be referred to as credit spread movements, can have significant impact on the capital and earnings of an institution. An example would be if an institution has invested in a bond for which they have paid a nominal value, which they will regain at maturity. The widening of the bond's credit spread would result in a gap increase between the nominal value paid and today's value of the bond. If the bond holder were to sell the bond today for market value, the sale would pose a direct impact on the institution's capital.

As mentioned, the credit spread movements are driven by two elements. These are formulated in the 2022 regulatory framework of the EBA [1] as:

- i) The changes of the *"market credit spread"*, or *"market price of credit risk"*, which represents the credit risk premium required by market participants for a given credit quality. It can be seen as the additional compensation that investors demand for assuming a greater risk of counterparty default;

- ii) The changes of the "*market liquidity spread*" which represents the liquidity premium that induces market appetite for investments and presence of willing buyers and sellers. It can be seen as the additional compensation that investors demand for holding an asset that is less liquid, meaning it is more difficult to buy or sell quickly.

Given the drivers of credit spreads, the CSRBB is said to capture the risk of an instrument's changing spread while assuming the same credit rating. Based on this, one conclusion is that credit spread risk is only present when instruments are traded on a marketplace where investor sentiment, i.e. investor demand for credit risk and liquidity premium, affect the market price of the instrument.

In order to understand credit spread risk, one can compare it to credit risk. The two types of risk are related but not equivalent. Credit risk is the risk of not being paid back on a debt, also called the risk of default [29]. Credit risk is usually measured by credit rating scales, such as Moody's or Standard and Poor's Global Ratings Scale. In contrast to credit risk, credit spread risk is not entirely related to an institutions probability of default, but, as previously mentioned, rather refers to the risk of loss in the bank's capital and earnings arising from movements in credit spread, while remaining in the same credit rating category. Two different bonds can have the same credit rating, but be exposed differently to credit spread movements.

Credit spread widening and tightening are not directly related to changes in creditworthiness of the borrower, but rather a consequence of general market view and risk factors such as investor's perception of the market, risk aversion, and demand for credit-risky securities [22]. A simple example would be investors, due to increased market uncertainty or worsening economic climate, turning to less risky securities, such as government bonds. As a widespread behaviour, this would in turn cause government bond prices to rise and the yield of more risky assets, such as corporate bonds, to fall. The widening of spread between the two bonds would indicate an increased uncertainty in market conditions, caused by factual change or market perception [12]. Credit spreads can thus be good indicators of current market conditions and investor sentiment.

2.2 Regulatory Environment

The European Banking Authority (EBA) is a regulatory body of the European Union (EU) and one of three supervisory authorities of the European System of Financial Supervision (ESFS). It is mandated to assess risks and vulnerabilities in the EU banking sector, providing prudential regulations and supervisory practices that contribute to a single set of EU rules for financial institutions (known as the European Single Rule book) in banking. The EBA is responsible for ensuring the efficient and consistent functioning of the EU banking sector, especially in periods of economic stress. In Sweden, the Financial Supervisory Authority (FSA), Finansinspektionen (FI) in Swedish, works to maintain stability of the financial system in accordance with EU standards and regulations, including those set forth by the EBA.

The Basel Committee on Banking Supervision (BCBS) is a group of members consisting of central banks and banking regulators across the world. Basel establishes international standards for banking regulation which are known as the Basel Accords. The EBA works closely with Basel and works actively with translation of the international standards into EU law, monitoring their compliance of EU banks as well as developing EU-specific regulations.

Until recently, banks have adapted their risk measurement tools and strategies to suit a low-for-long interest rate environment, guided by the EBA and Basel. However, in the current higher interest rate environment, banking supervisory entities have increased their regulatory pressure on institutions to enhance their assessment and monitoring of interest rate risk and credit spread risk [7]. This increased regulatory pressure aims to improve the quality and robustness of financial institutions' financial structures. In 2018, the EBA issued the framework EBA/GL/2018/02 "*Guidelines on the management of interest rate risk arising from non-trading book activities*". The guidelines were updated in October 2022 to incorporate new directives on IRRBB and CSRBB, as well as two sets of Regulatory Technical Standards (RTS). These three new regulatory frameworks aim to implement the Basel standards on IRRBB and CSRBB in the EU in response to recent developments in the economic environment. The 2022 guidelines place a greater emphasis on CSRBB as a separate risk category compared to the 2018 framework.

More precisely, the EBA 2022 regulatory framework presents four key specifications to the assessment and calculation of CSRBB:

- i) Institutions are required to assess CSRBB in all financial instruments, i.e assets, liabilities, and off-balance-sheet items, that are sensitive to credit spread risk;
- ii) All fair valued assets must be included in the CSRBB assessment at a minimum;
- iii) The exclusion of certain financial instruments in the assessment of CSRBB must be motivated;
- iv) Measurement of CSRBB must be performed in terms of Economic Value of Equity (EVE) and Net Interest Income (NII).

However, in addition to some specifications on the assessment and calculation of CSRBB, the EBA also emphasises the importance of institutions to develop and use their own assumptions and calculation methods. The EBA states that *"Institutions should develop and use their own assumptions and calculation methods for the assessment of CSRBB. The choice of measurement methodology should be adequate for the complexity of the bank itself"* [1].

This statement highlights that while the framework provides guidance, it still contains elements subject to ambiguity, requiring institutions to take on responsibility and apply their own methodologies as they investigate their CSRBB exposures.

In 2020, the FI introduced a standardised model for credit spread risk, aligned with the 2018 EBA guidelines. This model serves as a replacement for internal risk models used by banks, which calculates the credit spread risk exposure of the banks' transferable assets, i.e. excluding liabilities. The FI model incorporates pre-specified factors that reflect stress scenarios related to credit spreads, taking into account issuer types and their respective credit ratings. These stress factors are determined through a combination of qualitative evaluation and historical analysis, examining credit spread movements during the period from 2008 to 2018. Additionally, as an alternative to the standardised model, the FI proposes the utilisation of a VaR (Value at Risk) model, employing a 90-day holding period and a confidence level of 99 percent [4].

2.3 Measures of CSRBB

There are two groups of risk measures required as part of the regulatory framework of the EBA: one is the value-based Economic Value of Equity (EVE) and the other is the earnings-based Net Interest Income (NII) plus market value changes. These are

common metrics in standard asset and liability management and are based on changes in the net present value and expected future profitability of instruments sensitive to credit spread risk. Institutions should, according to the EBA, use both NII and EVE as complementary measures to quantify the credit spread risk over both short- and long-term horizons. For a more detailed explanation on how to calculate EVE and NII, the reader is referred to section 4.4.

The net present value of all asset and liability cash flows is taken into account in the EVE measure, which represents a form of long-term economic value measure. It is calculated over the remaining holding times of the instruments, i.e. until all banking book positions have run off. To estimate the potential loss of economic value of equity resulting from credit spread risk, i.e. the difference in EVE between a stressed and a base scenario, the net present values of the portfolio bonds under each scenario are aggregated.

The NII measure, on the other hand, represents the expected profitability over a specified future time-horizon and is based on the net interest income resulting from all interest-bearing assets and liabilities. Net interest income plus market value changes will take into account the market value changes of the instruments during the specified future time-horizon. In computing the NII, a constant balance sheet is assumed, meaning financial instruments that mature during the specified time period are replaced by new instruments with identical terms, such as nominal amount and repricing dates.

To simplify the mapping of cash flows from each bond in the portfolio, the BCBS recommends grouping all future cash flows into 19 predefined "time buckets", or repricing time intervals [9]. The cash flows within each time bucket are assumed to occur at the midpoint of the interval. Further details on mapping cash flow data and the specified time buckets for repricing cash flows can be found in section 4.2.

Chapter 3

Mathematical Framework

This chapter provides a theoretical overview of the mathematical methodologies and methods used in the project. The concepts presented will aid the reader to gain a deeper understanding of the model's development, implementation, and subsequent results presented in later chapters. The chapter covers essential topics such as the theory and derivations of the Forward Rate, the Nelson-Siegel-Svensson method, Value at Risk, and Expected Shortfall.

3.1 Forward Rate

A Zero Coupon Bond (ZCB) is a type of bond that does not provide any coupon payments until its maturity at time T . The price of a ZCB is denoted by $p(t, T)$, where t indicates the time at which the bond is contracted. At maturity, the ZCB pays the bondholder one unit of currency.

The forward rate, denoted as $F(t; S, T)$, relates to the time period $[S, T]$, and is agreed upon at time $t < S$ for a maturity $T > S$. The forward rate is used to calculate the future yield on a bond and can be expressed as a relationship between two ZCBs [11]:

$$e^{F(t; S, T)(T-S)} = \frac{p(t, S)}{p(t, T)}. \quad (3.1)$$

By solving equation (3.1) for the forward rate, we obtain

$$F(t; S, T) = -\frac{\ln p(t, T) - \ln p(t, S)}{T - S}. \quad (3.2)$$

An important remark is that as S approaches t , $F(t; S, T)$ converges to the spot rate $r(t, T)$, which is the continuously compounded interest rate used to price a ZCB:

$$p(t, T) = e^{-r(t, T)(T-t)}. \quad (3.3)$$

Equations (3.2) and (3.3) provide a relationship between forward rates and spot rates, which will be used to calculate the projected cash flows in the NII calculation of the portfolio (c.f. subsection 4.4.2).

3.2 The Nelson-Siegel-Svensson Model

3.2.1 Yield Curve Modelling

As introduced in section 2.1, in order to calculate credit spreads, two sets of yield data for each security: the yield of the risky security, e.g. a bond, and the yield of a benchmark (risk-free) security with the same maturity are necessary. The difference between these is the credit spread. However, since it is highly likely that necessary market data for specific maturities is unavailable, a method to estimate theoretical yields, or interest rates, needs to be applied. The term structure of interest rates, also known as the yield curve, is defined as the relationship between short- and long-term interest rates. The yield curve varies as a function of the years remaining to maturity, and can be illustrated by the blue and black graphs in figure 2.1.1.

In the scope of approximating yields, there are a number of common models widely used in the industry. In the comparative study of the ECB Statics Paper Series “*Yield curve modelling and a conceptual framework for estimating yield curves: evidence from the European Central Bank’s yield curves*”[3] there are two groups presented from which the zero-coupon and forward rates from observed bond data can be derived. The first group is known as spline-based methods, and the second parsimonious parametric models. Both groups are used by central and national banks of the EU, although the parametric Nelson-Siegel (NS) model and its extension by Svensson (Nelson and Siegel, 1987; Svensson, 1994) are more dominant. All yield curve models are based on different ideas of how the yield can be decomposed into several components, each of which describes a different aspect of the yield curve’s behaviour.

For the purpose of this project, the Nelson-Siegel-Svensson (NSS) technique has been chosen due to its parsimonious nature, meaning its ability to explain data with few and simple assumptions, and generally pre-eminent ability of term-structure forecasting compared to spline-based methods. The extended NSS method is considered more advantageous to the simplified NS model due to its inclusion of two additional function parameters, allowing the model to better capture the behaviour of the yield curve at longer maturities. These long-term level parameters is particularly useful as it allows for better predictions for maturities beyond the data range, meaning for longer maturities than what there is data for, something that other methods fail to do [19].

The aim of yield curve modelling is essentially to fit a set of discrete empirical yields to a pre-determined mathematical function. To interpolate between the available data points, various optimisation techniques can be used, which will be further discussed in subsection 3.2.3.

3.2.2 Derivation

As the NS and NSS models are both variations of exponential polynomials, they are generally explored together. The NS model is motivated by the relationship between spot rates and forward rates. It is argued that if spot rates are generated by a differential equation, then forward rates, being forecasts, will be the solution to this equation.

The instantaneous forward rate, denoted as in equation (3.3), is under the NS model given by:

$$r(t, T) = \beta_0 + \beta_1 e^{\frac{-T}{\tau}} + \beta_2 \left[\left(\frac{T}{\tau} \right) e^{\frac{-T}{\tau}} \right]. \quad (3.4)$$

This form assumes that the yield curve can be represented as a linear combination of three factors: the level factor, the slope factor, and the curvature factor, denoted in equation (3.4) by β_0 , β_1 , and β_2 , respectively. The non-linear parameter, τ , dictate the location and flexibility of the curvature by controlling the the speed of convergence towards the long-term average [25].

The zero yield curve, $R(t, T)$, can be obtained by integrating the spot rates using the relationship:

$$R(t, T) = \frac{1}{T} \int_0^T r(t, x) dx.$$

Finally, the zero yield curve is then given by:

$$R(t, T) = \beta_0 + (\beta_1 + \beta_2) \left[1 - e^{-\frac{T}{\tau}} \right] \frac{T}{\tau} - \beta_2 e^{-\frac{T}{\tau}}$$

or, by collecting terms,

$$R(t, T) = \beta_0 + \beta_1 \left[1 - e^{-\frac{T}{\tau}} \right] \frac{T}{\tau} + \beta_2 \left[\left(1 - e^{-\frac{T}{\tau}} \right) \frac{T}{\tau} - e^{-\frac{T}{\tau}} \right]. \quad (3.5)$$

The NS model can generate several shapes of the yield curve, including upward sloping, downward sloping, and hump shaped with one minimum and maximum. The different shapes are generated by altering the parameters in equation (3.5).

The NSS model builds upon the NS model, aiming to enhance flexibility and improve the model's fit to yield data. By introducing two additional parameters (β_3 and τ_2) in an extra rate term, the NSS model allows for the inclusion of a second hump shape in the yield curve. This extended approach is considered more advantageous than the NS model since it can generate a yield curve with two humps instead of one. Although rare, a yield curve with two humps can occur in reality, which is why the extended model is considered more suitable for practical application.

Under the NSS model, the instantaneous forward rate, or spot rate, is given by:

$$r(t, T) = \beta_0 + \beta_1 e^{-\frac{T}{\tau_1}} + \beta_2 \left[\left(\frac{T}{\tau_1} \right) e^{-\frac{T}{\tau_1}} \right] + \beta_3 \frac{T}{\tau_2} e^{-\frac{T}{\tau_2}} \quad (3.6)$$

and the zero yield curve

$$\begin{aligned} R(t, T) = & \beta_0 + \beta_1 \left[1 - e^{-\frac{T}{\tau_1}} \right] \frac{T}{\tau_1} + \beta_2 \left[\left(1 - e^{-\frac{T}{\tau_1}} \right) \frac{T}{\tau_1} - e^{-\frac{T}{\tau_1}} \right] \\ & + \beta_3 \left[\left(1 - e^{-\frac{T}{\tau_2}} \right) \frac{T}{\tau_2} - e^{-\frac{T}{\tau_2}} \right]. \end{aligned} \quad (3.7)$$

To illustrate the fit of both the NS curve and the NSS curve, the two methods can be applied to the same yield data and plotted in the same diagram. Figure 3.2.1 shows one example of their respective yield curve estimations.

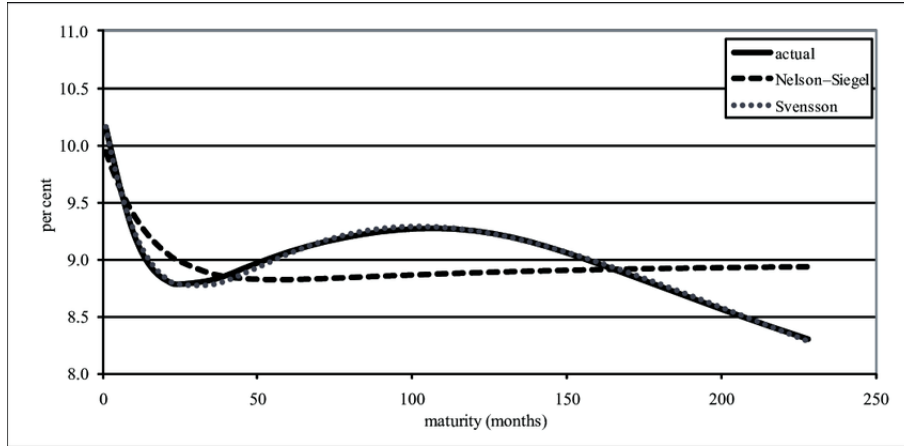


Figure 3.2.1: Nelson-Siegel fit versus Svensson fit of the yield curve [20].

3.2.3 Parameter Calibration and Estimation

The linear and non-linear parameters of the NSS model can be estimated simultaneously using non-linear optimisation techniques, such as non-linear least squares (NLS), which minimises the quadratic difference between estimated spot rates and observed market rates. For several days of yield data (for $t = 1, \dots, N$), we solve equation (3.7) separately each day, resulting in N sets of parameters denoted by the subscript t . The non-linear optimisation problem to be solved can then be written as:

$$\min_{\lambda, \beta} \sum (\hat{r}(t, T) - r^M(t, T))^2 \quad \forall t,$$

where $\hat{r}(t, T)$ denotes the estimated spot rate, and $r^M(t, T)$ the market spot rate for maturity T and day t .

Alternatively, for greater simplicity and little loss-of-fit, as suggested by Diebold and Li [13], the non-linear elements of λ could instead be fixed. This can greatly simplify the iterative calibration procedure of the least squares method by only considering the linear parameters, and thus reducing the model to a straightforward linear regression. The parameters can then be estimated using an ordinary (linear) least squares (OLS) method instead of the NLS.

However, potentially challenging non-linear optimisation procedures can be avoided by simply choosing appropriate numerical computing tools with robust mathematical calculation abilities. For the sake of accuracy, and based on the numerical environment

used for this project, a non-linear least squares method was implemented in the Matlab environment of MathWorks using a built-in function for non-linear data-fitting. To further ensure stability and accuracy of fit, the resulting yield curves of the fitted model were plotted and inspected. The resulting yield curves of the NSS model method can be presented and illustrated in subsection 5.1.2.

3.3 Value at Risk

3.3.1 Derivation

A portfolio's VaR at a given level $p \in (0, 1)$ can be defined using the equation:

$$\text{VaR}_p(X) = \min \{m : P(mR_0 + X < 0) \leq p\},$$

where X is the portfolio's value at time 1 and R_0 is the percentage return of a risk-free asset, e.g. the yield of a government ZCB. This equation can also be written as:

$$\text{VaR}_p(X) = \min\{m : P(-X/R_0 \leq m) \geq 1 - p\}, \quad (3.8)$$

which represents the minimum loss amount m such that the probability of the loss being at most m is at least $1 - p$.

Here, we introduce the variables V_0 and V_1 to represent the value of the portfolio at times 0 and 1. If we compare the scenarios of investing V_0 in a risk-free asset with return R_0 and investing the same amount in a risky portfolio at time 0, we obtain the net value $X = V_1 - V_0R_0$ at time 1. Additionally, by treating V_0 as a liability, we obtain L , which can be interpreted as the discounted loss at time 1 and is given by

$$L = -\frac{X}{R_0} = V_0 - \frac{V_1}{R_0}.$$

By exchanging variables in equation (3.8) we obtain

$$\text{VaR}_p(X) = \min\{m : P(L \leq m) \geq 1 - p\}, \quad (3.9)$$

which can be interpreted as the probability of the discounted loss L being at most m

is at least $1 - p$. In other words, VaR can be viewed as the $(1 - p)$ -quantile of L [15].

We can rewrite (3.9), to obtain the quantile function of the distribution F_L of the random variable L as:

$$\text{VaR}_p(X) = \min \{m : F_L(m) \geq 1 - p\} = F_L^{-1}(1 - p).$$

3.3.2 Risk Measurement

The following subsection provides an introduction to the general principles of risk measurement and the context of VaR.

In order to analyse the risk of a portfolio, it is convenient to investigate the probability distribution of the future portfolio value in terms of a single comparable value. Thus, one may use a risk measurement function, $\rho(X)$, where

- X = possible future values of the portfolio, at time 1
- $\rho(X)$ = function that assigns real numbers (or $+\infty$) to each future portfolio value, representing the risk measurement of X .

In order to obtain an appropriate risk measurement function $\rho(X)$, a number of axioms, or properties, have been suggested. A selection of the axioms are presented below, where X_1 and X_2 are two equally distributed future portfolio values for which the risk measure can assign different values and R_0 is the percentage return, between time 0 and 1, of a risk-free asset, e.g. a government ZCB [15]:

Translation invariance

This property says that adding cash reduces risk, i.e. when cash is added to the portfolio, the risk will reduce with the same amount.

$$\rho(X + cR_0) = \rho(X) - c$$

Monotonicity

This property says that if a portfolio's value always is greater than another portfolio's value at time 1, then the first portfolio must be less risky than the second

portfolio.

$$\text{if } X_2 \leq X_1 \text{ then } \rho(X_1) \leq \rho(X_2)$$

Positive homogeneity

This property says that risk is proportional to size, i.e. if the value of the portfolio at time 1 is increased by a factor, then the risk of the portfolio should be increased by the same factor.

$$\rho(\lambda X) = \lambda \rho(X) \text{ for all } \lambda \geq 0$$

In the context of risk management in financial institutions, VaR is an example of a risk measurement function. One may interpret $\rho(X)$ as the minimal capital required to be invested in a buffer at time 0 in order to provide sufficient protection against undesirable events at time 1, thus the maximum loss not to be exceeded, i.e. the VaR [15]. As illustrated below, the VaR fulfills translation invariance, monotonicity and positive homogeneity. The reader is referred to subsection 3.3.1 for relevant variables in equations below.

Translation invariance

$$\begin{aligned} F_{a+L}^{-1}(1-p) &= \min\{m : P(a+L \leq m) \geq 1-p\} \\ &= \min\{m : F_L(m-a) \geq 1-p\} \\ m' = m - a &\rightarrow m = m' + a \\ F_{a+L}^{-1}(1-p) &= \min\{m' + a : F_L(m') \geq 1-p\} \\ &= a + \min\{m' : F_L(m') \geq 1-p\} \\ &= a + F_L^{-1}(1-p) \end{aligned}$$

Positive homogeneity

$$\begin{aligned}
 F_{bL}^{-1}(1-p) &= \min\{m : P(bL \leq m) \geq 1-p\} \\
 &= \min\left\{m : F_L\left(\frac{m}{b}\right) \geq 1-p\right\} \\
 m' = \frac{m}{b} &\Rightarrow m = m'b \\
 F_{bL}^{-1}(1-p) &= \min\{m'b : F_L(m') \geq 1-p\} \\
 &= b \cdot \min\{m' : F_L(m') \geq 1-p\} \\
 &= b \cdot F_L^{-1}(1-p)
 \end{aligned}$$

Monotonicity

Assume that $L_2 \leq L_1$, then

$$\begin{aligned}
 P(L_1 \leq m) &\leq P(L_2 \leq m) \\
 F_{L_1}(m) &\geq F_{L_2}(m) \\
 \min\{m : F_{L_1}(m) \geq 1-p\} &\geq \min\{m : F_{L_2}(m) \geq 1-p\}
 \end{aligned}$$

3.3.3 Comparing VaR methods

The following subsection will present a comparison of various methods for obtaining the VaR measure, along with a discussion of their respective strengths and limitations.

The basic idea of VaR methods is to generate scenarios of a variable that is driven by a probability distribution. The scenarios are then used to measure the maximum loss not to be exceeded during a given time period. The parameter of interest could for example be the price of a stock or credit spreads. There are three VaR methods commonly used in the financial industry: the historical simulation method, Variance-Covariance method, and Monte Carlo method.

The historical simulation method is based on the assumption that historical patterns indicate future scenarios. It is a method that is easy to use and implement if there exists historical data on the relevant variable. A drawback of the historical simulation method is that it assumes that only historical events can happen again and do not account for new, and possibly more extreme scenarios.

The Variance-Covariance method assumes the variable to be driven by a normal distribution. The standard deviation of the variable is calculated, and in order to calculate the VaR at a certain confidence level, the standard deviation is multiplied with a known factor [5]. The method is useful due to its simplicity when the data set is known to be normally distributed. However, assuming a perfectly normal distribution could be a drawback, as it does not take into account the potential kurtosis or skewness of the data. Additionally, the method assumes constant covariances and correlations which is not necessarily true in reality [6].

Monte Carlo methods are based on creating a large number of simulations of the variable based on a defined probability distribution. Monte Carlo methods are advantageous from a risk modeling perspective as it can, in contrast to historical simulation methods, create entirely new scenarios. The method is useful when handling more complex stochastic models such as *GARCH*. However, compared to previously mentioned VaR methods, the method can be rather computationally intensive.

The historical simulation method was chosen for the model of this paper, due to its simplicity and widespread usage in the financial industry.

3.4 Expected Shortfall

Expected Shortfall (ES) is a coherent risk measure. In addition to translation invariance, monotonicity, and positive homogeneity, the risk measure also always fulfills subadditivity, which is a property saying that portfolio diversification reduces the risk of the portfolio, i.e. the risk of a portfolio holding two assets simultaneously is less or equal to the risk in holding them separately. ES is widely used in the financial industry, as it calculates the expected loss in the scenario of extreme events when the VaR cutoff point is crossed. A drawback of VaR is that the risk measure ignores the left tail, meaning the values beyond the cutoff point are not taken into consideration. The ES calculates the average VaR below the cutoff point, thus measuring the risk of severe loss events with small probabilities.

The expected shortfall at level p is defined by:

$$ES_p(X) = \frac{1}{p} \int_{1-p}^1 F_L^{-1}(u) du$$

or

$$ES_p(X) = \frac{1}{p} \int_0^p \text{VaR}_u(X) du. \quad (3.10)$$

If X has a continuous distribution function, we can rewrite equation (3.10) with $L = -X/R_0$ as:

$$ES_p(X) = E[L \mid L \geq F^{-1}(1-p)] = E[L \mid L \geq \text{VaR}_p(X)],$$

which expresses the expected shortfall as the expected value of the discounted loss, conditional on the loss being greater than the VaR cutoff point [15].

Chapter 4

Methodology

This chapter describes the application of presented concepts in the final model, as well as a step-by-step method for arriving at the results, seen in figure 4.0.1. Firstly, the chapter introduces the studied balance sheet, which forms the basis for the findings of this thesis. Supporting the calculations of CSRBB, the mapping of cash flow data is described, as well as the historical data used in the model. Moving on, the EVE and NII assessment and calculations are presented in more detail, and finally, the process of backtesting the model is explained.

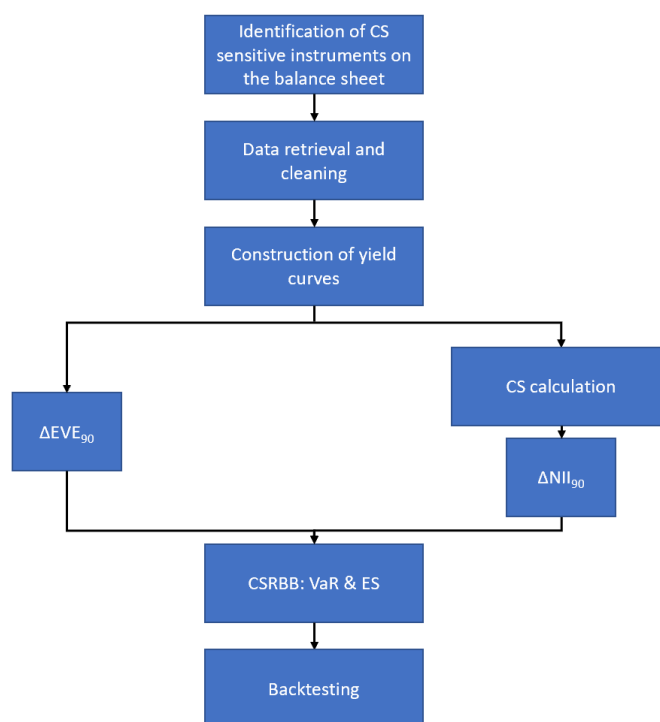


Figure 4.0.1: Flow chart illustrating the steps of the model.

4.0.1 Step-by-Step Method Overview

- 1) Identify all credit spread sensitive instruments on the balance sheet and slot each cash flow into its respective time bucket. The cash flow mapping is described in section 4.2 and the time buckets are illustrated in figure 4.2.1;
- 2) Categorise each credit spread sensitive instrument according to the FI issuer categories, found in subsection 4.3.2. Retrieve historical market yield data for each category, as well as for the risk-free benchmark bond, which in this model is given by government bonds. Specifications on the data used for the modelling is found in subsections 4.3.1 and 4.3.2;
- 3) Use the Nelson-Siegel-Svensson method to construct daily yield curves for all instruments, risky and risk-free;
- 4) For the NII calculations, find the daily historic credit spreads by subtracting the yield of the government bond from the yield of each risky bond. This is done for each day of the collected historical data;
- 5) Calculate the EVE and NII for a base and a stressed scenario and obtain ΔEVE and ΔNII . Lastly, obtain the rolling 90-day difference in ΔEVE and ΔNII (ΔEVE_{90} and ΔNII_{90}) for all past values;
- 6) Calculate the CSRBB by sorting, in descending order, all historical ΔEVE_{90} and ΔNII_{90} , and obtaining the $\text{VaR}_{0.99}$ and $\text{ES}_{0.99}$ values;
- 7) Perform backtesting on the resulting VaR and ES measures for the full previous year of the balance sheet date.

4.1 Balance Sheet

Based on the EBA definitions and guidelines discussed in chapter 2, the credit sensitive instruments on the balance sheet need to be identified. As stated in section 2.1, CSRBB captures the credit risk and liquidity premium induced by sentiment of market participants. Thus, it is important to understand the underlying prerequisite of CSRBB, that is, the possibility to trade the instruments on the market.

The studied balance sheet is constructed based on a collection of balance sheets from multiple institutions. The instruments collected in figure 4.1.1 are considered to be

subject to credit spread risk and have therefore been included in the model. In-detail discussions of respective category are provided in subsections 4.4.1 and 4.4.2. The fictional banking book includes assets and liabilities of three different currencies: Swedish krona (SEK), Euro (EUR), and Norwegian krone (NOK).

	Assets	Liabilities
Bonds	Floating Rate Bonds	Floating Rate Bonds
	Zero Coupon Bonds	Perpetual bond (floating interest rate)
	Fixed Coupon Bonds	
Derivatives	Interest Rate Swap, floating leg (floating interest rate)	Interest Rate Swap, fixed leg (fixed interest rate)
Lending	Mortgages (floating interest rate)	

Figure 4.1.1: Credit spread sensitive instruments on the balance sheet.

The size of the cash flows generated by the balance sheet positions are illustrated in figure 5.1.1. The timing of these cash flows have been categorised according to the time bucket method explained in section 4.2.

4.2 Mapping of Cash Flow Data

In order to simplify the calculations of EVE and NII, the EBA has suggested a time bucket approach, or "cash flow bucketing" [16]. This involves slotting cash flows into pre-specified time buckets determined by the time at which they are expected to be received or paid out. The time of each cash flow is denoted t^{CF} , and can for each bucket be found in figure 4.2.1. The midpoint of each time bucket will represent the new repricing time-point for all cash flows allocated in that time bucket. As illustrated, there are a total of 19 buckets representing cash flows within the short-, medium- and long-term horizons. The resulting projected cash flows of the studied balance sheet, allocated to its specific time bucket, can be found in figure 5.1.1.

	Time bucket intervals (M: months; Y: years)							
Short-term rates	Overnight (0.0028Y)	O/N < t^{CF} $\leq 1M$ (0.0417Y)	1M < t^{CF} $\leq 3M$ (0.1667Y)	3M < t^{CF} $\leq 6M$ (0.375Y)	6M < t^{CF} $\leq 9M$ (0.625Y)	9M < t^{CF} $\leq 1Y$ (0.875Y)	1Y < t^{CF} $\leq 1.5Y$ (1.25Y)	1.5Y < t^{CF} $\leq 2Y$ (1.75Y)
Medium-term rates	2Y < t^{CF} $\leq 3Y$ (2.5Y)	3Y < t^{CF} $\leq 4Y$ (3.5Y)	4Y < t^{CF} $\leq 5Y$ (4.5Y)	5Y < t^{CF} $\leq 6Y$ (5.5Y)	6Y < t^{CF} $\leq 7Y$ (6.5Y)			
Long-term rates	7Y < t^{CF} $\leq 8Y$ (7.5Y)	8Y < t^{CF} $\leq 9Y$ (8.5Y)	9Y < t^{CF} $\leq 10Y$ (9.5Y)	10Y < t^{CF} $\leq 15Y$ (12.5Y)	15Y < t^{CF} $\leq 20Y$ (17.5Y)	$t^{CF} > 20Y$ (25Y)		

Figure 4.2.1: Time bucket intervals for repricing cash flows.

4.3 Credit Spread Calculations

4.3.1 Model Data

There are few but important regulatory notes concerning the data sourcing of the credit spread risk model. According to the EBA, credit spread risk measures used for the internal model should be based on reliable market and internal data [1]. Following this, the procurement of market data has been made using the Eikon specialist database.

In the FI standardised model on credit spread risk, the time-period for which credit spreads was considered lapsed between the year of 2008 and 2018. The 10-year interval was chosen due to its inclusion of two longer financially stressed time-periods, as shorter time periods may exclude historical data related to financial stress and, as a result, underestimate risk capital margins. Similarly, in the general model requirements of the FI when developing and using an internal VaR model, FI asks institutions to base their calculations on historical data with multiple stressful periods [4].

To fulfill the requirement of historical data including severe stresses, the choice of data for this thesis has been chosen ten years prior to the date of the balance sheet. This is to say the data-set covers financial crises such as the Eurozone debt crisis of 2012/13, the Covid-19 crisis of 2020, and the Russian invasion of Ukraine of 2022/23 [31]. From the starting date of data until the balance sheet date, daily market data of each bond in holding is obtained.

4.3.2 Risk Categories

Since the model is built based on historical data, and the access to relevant data is in general limited, it is necessary to use generic historical yield data. By generic data, we refer to yield data which represent historical fluctuations of the holdings in the given portfolio. The FI suggests splitting the portfolio holdings into six different categories for its standardised approach to assess credit spread risk, under which the given portfolio of this thesis falls into four of these [4]:

- States and Supranational Organisations;
- Municipalities and Regions;
- Instruments issued by credit institutions;
- Covered Mortgage Bonds.

Additionally, in order to find the credit spread of the portfolio holdings, a presumed risk-free benchmark is needed. In this paper, government bonds have been used for the purpose of constructing a risk-free benchmark rate. The decision to use government bonds is based on industry standards for a risk-free reference rate, as well as the nature of our bond portfolio [18]. The reader is referred to subsection 5.1.2 for a more detailed discussion on the selection process of the benchmark rate. A specifying table of the yield data used, as well as modifications applied to the data, can be found in appended section A.1.

4.3.3 Yield Curve Construction

After slotting the expected cash flows of the balance sheet positions into their respective time bucket, as described in section 4.2, corresponding yield data for the time bucket-specific maturities need to be obtained. The yield data for all financial instruments of the portfolio, as well as for the presumably risk-free benchmark rate, will be used in the NSS method to obtain yield curves, from which any maturity yield can be obtained.

Plotting the resulting yield curves serve as a critical tool for evaluating their stability and accuracy. Since the shape of curves can have a significant impact on the resulting risk estimations, it is essential to ensure that the yields used in the model are both realistic and consistent over time. By analysing the historical yield data and fitting

to smooth curves using the NSS method, one can examine the degree to which the yield curves deviate from their trend and thus assess their reliability for risk estimation purposes.

Stable yield predictions refer to yield curves that are relatively consistent and predictable over a given period of time. In the case of the yield curves presented in this thesis, stable yield predictions up to maturity twenty-five years mean that one can reasonably forecast how the yields of bonds with maturities up to 25 years will behave. It is important to note, however, that yield curve predictions are subject to a number of uncertainties and risks. Factors such as changes in interest rates, economic conditions, and shift in investor sentiment can all impact the behaviour of bond yields, and thus the accuracy of yield curve predictions.

After sets of daily historical yields have been fitted to their respective yield curve, the resulting credit spread (CS) can be found for each day t and maturity m . It is calculated as the yield of the risk-free bond RF subtracted from the risky bond R as given by

$$CS(t, m) = R(t, m) - RF(t, m),$$

which will later on be used in the calculation of the NII measure in subsection 4.4.2.

The number of historical daily credit spreads will thus be the same as the number of historical dates of yield data used. The resulting historical daily credit spread are illustrated in the following section.

4.3.4 Historical Credit Spreads

The CSRBB model is based on data of historical credit spreads, which are presented in figures 4.3.1 to 4.3.3. The peaks and troughs of the spreads are reflecting significant financial events and crises affecting the global economy over the past ten years. As noted earlier in the thesis, larger credit spreads indicate market instability, while smaller credit spreads suggest market security and stability. As discussed in subsection 4.3.1, the model is based on historical data from the past ten years, which includes at least two prolonged periods of financial stress.

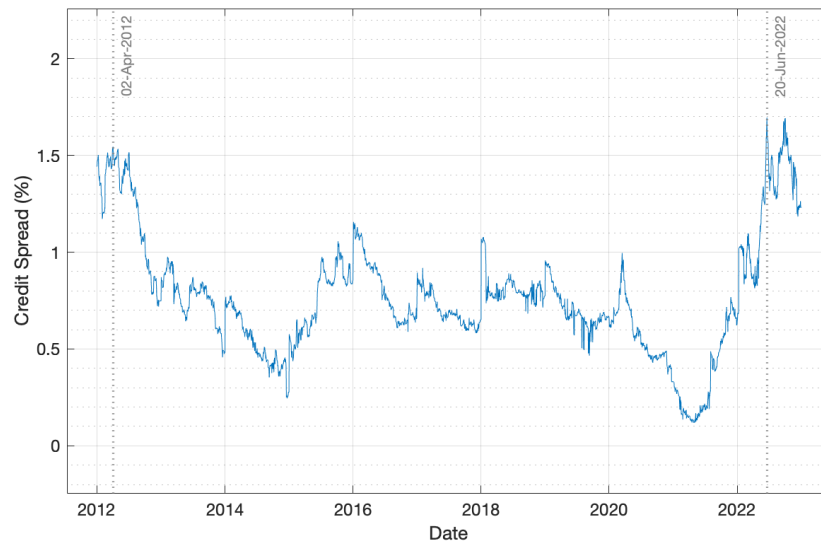


Figure 4.3.1: Historical credit spreads between Swedish government bonds and mortgage bonds for maturity five years.

Figure 4.3.1 displays the credit spread between Swedish government and mortgage bonds for a maturity of five years, which is part of the data used in the model. The dates for the two largest credit spreads during two separate time periods are marked in the figure. To ensure that the data used in the model is "stressful enough" and provides sufficiently risk-conservative results, we can examine a larger historical time-frame of the past twenty years. We can then compare the size of the two largest credit spread periods to all observed stresses since the early 2000s. As showed in figure 4.3.2, only two additional periods of larger financial stresses are observed, dating around 2009 and early 2012, indicating that the data used in the model is sufficiently stressed.

Similar to the comparison made above, we can also examine the credit spread history of the floating rate-sensitive instruments. In this case, the spread between government bonds and bonds of the underlying floating interest rate captures their credit spread sensitivity. Figure 4.3.3 displays the credit spread history since the beginning of the year 2000 between Swedish government bonds and the Stibor OIS for a maturity of ten years. The dotted vertical line in the figure indicates the beginning of 2013, which is the starting point of the data included in the model.

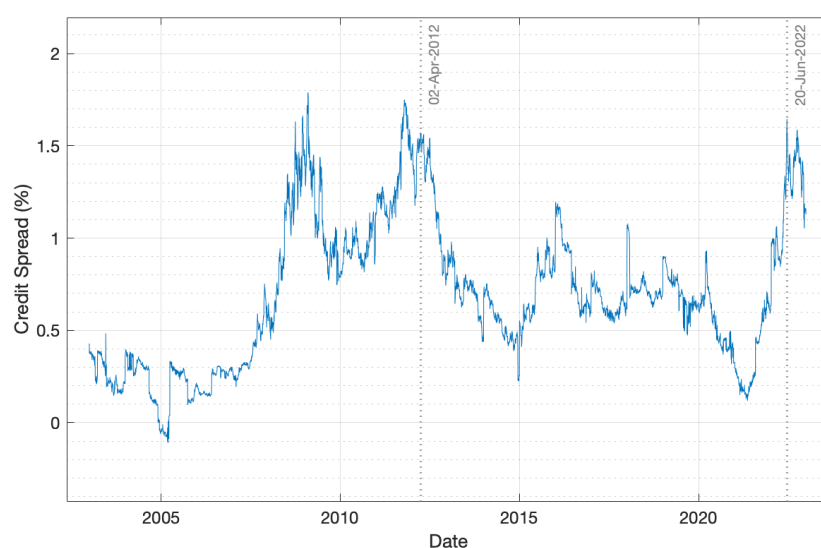


Figure 4.3.2: Historical credit spreads between Swedish government bonds and mortgage bonds for maturity five years.

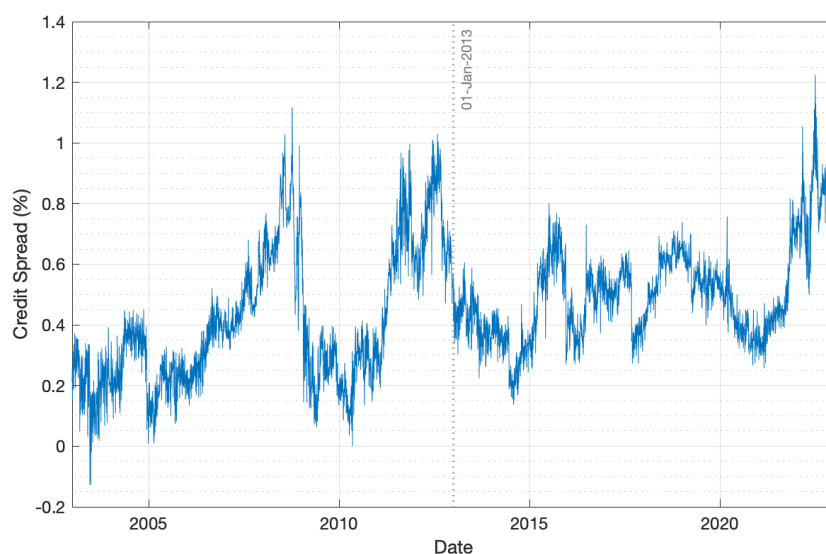


Figure 4.3.3: Historical credit spreads between Swedish government bonds and Stibor OIS for maturity ten years.

As discussed in chapter 2, the credit spread is sensitive to the yields of the two compared instruments. It is therefore crucial to use representative yield curves in order to obtain relevant credit spread values. Subsection 5.1.2 provides illustrations of the yield curves used in the model.

4.4 Value at Risk Calculations

Regarding the type of VaR method applied, a historical simulation VaR method was chosen due to its simplicity and widespread usage in the financial industry. The HSVaR method provides the risk of loss based on historical credit spread movements of each financial instrument on the balance sheet.

Following in this section, the reader will be presented the calculations used for the CSRBB risk measures; EVE and NII. In addition, their respective approach and interpretation for the asset and liability side of the balance sheet will be discussed.

4.4.1 EVE

EVE is the net present value of financial instruments, i.e. the aggregated value of the financial instruments if they were to be sold on the market today. The interpretation of EVE differs in regard to the asset and liability side of the balance sheet, which will be discussed in the following passages.

Assets

The question discussed in the following passage: what asset instruments on the balance sheet should be included in the calculation of EVE?

In calculating the EVE, fixed coupon bonds and zero coupon bonds are of interest, since it is generally always possible for the bond investor to sell these bonds on the market. Market price movements can be induced by widening and tightening of credit spreads, and therefore, they are subject to credit spread risk.

However, in the case of floating interest rate instruments, it is not as clear. The floating interest rate bonds, the mortgage bonds, as well as the floating leg of the interest rate derivatives on the balance sheet, all follow the floating Stockholm Interbank Offered Rate (Stibor) or the Norwegian Interbank Offered Rate (Nibor). These benchmark rates are commonly used as benchmarks in financial contracts. Stibor and Nibor are calculated on a daily basis and are based on the mean of interest rates offered by large Swedish and Norwegian banks for lending capital short term without security [21]. Stibor and Nibor are both subject to the influence of market sentiment, and therefore, floating rate instruments are considered to be subject to credit spread risk.

The interest rate, or coupon, of the floating interest rate instruments changes periodically. The date of change in floating interest rate is known as the instruments' repricing dates. Therefore, the interest rates can be considered fixed until the next upcoming repricing date. In the studied balance sheet, the floating instruments all follow Stibor 3M or Nibor 3M, meaning the interest rates are known and fixed for a quarter of a year. Following this, the floating interest rate instruments can be seen as fixed coupon bonds with payments at the next repricing date and at maturity. Each instrument will have different repricing dates based on their issue date.

For the EVE calculations, the cash flows from the first coupon and the notional value are therefore used. As the underlying benchmark interest rate is unknown up until the next repricing date, no additional cash flow of the floating interest rate was included.

Liabilities

The question discussed in the following passage: what liability instruments on the balance sheet should be included in the calculation of EVE?

What liability instruments to be included in the scope of the EVE risk measure is not entirely obvious. There are no regulatory guidelines providing suggestions on the type of liability instruments to include or how these should be handled in terms of risk calculations. In other words, there is a lot of room for interpretation.

It is debatable whether liabilities should be included in the calculation of the EVE risk measure. In the banking book, the Held-to-Maturity (HTM) instruments are reported at book value, i.e. their amortised cost, which is the historical price adjusted for any amortisation or accretion of discount or premium. Thus, the bond value is not recognised at fair value, i.e. the market value of the instrument, on the balance sheet. Therefore, any change in market value of the instrument induced by change in market sentiment would not be recognised, except if it were to be categorised as Available-for-Sale (AFS) and actually be sold.

It is generally not possible to go to a market place and sell liability instruments, since the bond issuer is legally obliged to pay the bond investor the nominal value at maturity, including coupons. For example, when having a bond as an asset, we are able to sell it whenever. However, having a bond as a liability, i.e. having issued a

bond and therefore obliged to pay the nominal value, we are not able to sell it however we please. Therefore, we are not introducing any market sentiment in the value of our bond. Evidently, it is possible for the bond investor to sell the bond on the secondary market, however, since the nominal value and the coupons are fixed, the bond issuer would not be affected.

One might consider the floating interest liabilities on the balance sheet to be subject to credit spread risk, since the issuer have to pay a floating interest rate that changes periodically, and is therefore influenced by market sentiment. However, as mentioned in subsection 4.4.1, when calculating the EVE, the floating interest rate is known only until the first repricing date, and therefore only the first coupon and the nominal value are slotted, as it up until this date can be considered a fixed coupon bond.

There are some possible scenarios where liabilities could be considered credit spread sensitive in terms of EVE. One scenario would be if the bond issuer decides to re-emit a bond with new contracted terms, where for example the yield has been adjusted due to changes in market conditions. The same argument can be applied in the case of the perpetual bond, since at the call date, the issuer has the opportunity to either buy back its bond for an agreed price, or must increase the interest rates paid to the bond-investor. Following these arguments, as well as unavailability of the contractual terms of the instruments of the studied balance sheet, no liabilities were included in the calculation of EVE in the risk model.

Calculation

To calculate the net present value of all cash flows, all net notional repricing cash flows are for each currency c , and time bucket $i \in (1, 2, \dots, 19)$, denoted $CF_{i,c}$, discounted to their present value using a continuously compounded discount factor

$$DF_{i,c,t} = e^{-R_{i,c}(t, m_i) \cdot (m_i - t_0)}. \quad (4.1)$$

Here, $R_{i,c}(t, m_i)$ is the interest rate at time t for maturity m_i and currency c , and discounted to time t_0 . A notional repricing cash flow is defined as any repayment of the principal, any repricing of principal, and any interest payment on a tranche of principal that has not yet been repaid or repriced [16].

The discounted cash flows can then be aggregated as

$$EVE_{stressed,c,t} = \sum_{i=1}^{19} CF_{i,c} \cdot DF_{i,c,t},$$

which provides the EVE of the stressed scenario.

In order to obtain the credit spread risk, the EVE in the base scenario is subtracted from $EVE_{stressed}$, where the presumably risk-free rate is used for the discount factor in the base scenario, and the rate of the risky bond is used for the discount factor in the stressed scenario:

$$\Delta EVE_{c,t} = EVE_{stressed,c,t} - EVE_{base,c,t}.$$

To obtain the total ΔEVE_t of all currencies on day t , all values have been converted to SEK, and then aggregated by

$$\Delta EVE_t = \sum_{c=1}^3 \Delta EVE_{c,t}.$$

In order to comply with FI's suggestion on introducing a 90 day holding period (c.f. section 2.2), we calculate the 90 day-rolling difference in ΔEVE_t , thus capturing the change in ΔEVE during 90 days. This is done by subtracting the value of ΔEVE on day 0 from the value of ΔEVE on day 90. By using the rolling 90 day difference, we ensure to cover all historic credit spread movements. While this may introduce some correlation between data, the capturing of all historical events, and thus obtaining a more conservative risk model, is considered to be of great importance. This operation results in several data points of what the change in EVE, induced by credit spreads, would be if we were to sell the portfolio on the market in 90 days.

Lastly, the $\text{VaR}_{0,99}$ is calculated in order to obtain the maximum loss not to be exceeded in 90 days, where ΔEVE_{90} denotes the 90 day rolling difference in ΔEVE for all currencies combined:

$$\text{VaR}_{0,99}(\Delta EVE_{90}) = \min \{m : P(\Delta EVE_{90} \leq m) \leq 0,01\}.$$

This expression represents the minimum value of m such that the probability of ΔEVE_{90} being less than or equal to m is at most 1%.

4.4.2 NII

When calculating the net interest income, we are interested in all the interest income and interest expenses during a limited time horizon of one year, as suggested by EBA. Since we are interested in the change in earnings, as a result of change in credit spreads of the financial instruments, we are only interested in the instruments for which the interest rate changes during the time period. As mentioned in section 2.3, in the calculation of NII, the balance sheet is assumed to be constant, meaning that when a financial instrument matures, it is replaced by a new one with identical terms, such as nominal value and repricing dates.

Floating interest rate bonds are of high importance when calculating the NII. Since they follow a floating benchmark rate, the interest rates are periodically adjusted based on the offered interest rates amongst banks, as mentioned in subsection 4.4.1, and are therefore considered to be subject to credit spread risk.

Firstly, all net notional cash flows for each respective currency are allocated to a time bucket. In order to project the interest earnings, we need to project the interest rates that are active during different points in time in the future. Therefore, the forward rates for time periods between time bucket midpoints are calculated. The forward rates are calculated by using the relationship between forward rates and spot rates, provided by equations (3.2) and (3.3), which result in the following formula:

$$F_t(t; T_k, T_{k+1}) = \frac{r_t(t, T_{k+1})(T_{k+1} - t) - r_t(t, T_k)(T_k - t)}{T_{k+1} - T_k}. \quad (4.2)$$

Here, T_k and T_{k+1} denote the midpoints of two following time buckets, i.e. repricing dates.

However, we are interested in the credit spread effect on NII and therefore apply the credit spread in equation (4.2). By assuming the forward rate for the time period between two repricing dates, T_k and T_{k+1} , to be contracted at time $t = 0$, the change in NII caused by the change in the credit spread is obtained by

$$\begin{aligned}
 \Delta NII_{c,t} &= NII_{stressed,c,t} - NII_{base,c,t} \\
 &= \sum_{i=1}^{19} CF_{i,c} \left(\left(F_{i,t}(0; T_{k+1}, T_k) - F_{i,t}^{rf}(0; T_{k+1}, T_k) \right) \right. \\
 &= \sum_{i=1}^{19} CF_{i,c} \left(\frac{(r_t(0; T_{k+1}) - rf_t(0; T_{k+1})) T_{k+1} - (r_t(0; T_k) - rf_t(0; T_k)) T_k}{T_{k+1} - T_k} \right) \quad (4.3) \\
 &= \sum_{i=1}^{19} CF_{i,c} \left(\frac{CS_t(0; T_{k+1}) T_{k+1} - CS_t(0; T_k) T_k}{T_{k+1} - T_k} \right),
 \end{aligned}$$

where r denotes the rate of the risky bond, and rf denotes the rate of the risk-free bond. The resulting $\Delta NII_{c,t}$ denotes the projected change in net interest income for currency c at time point t , and $F_i(t; T_k, T_{k+1})$ and $F_i^{rf}(t; T_k, T_{k+1})$ the effective yield of a risky bond and risk-free bond respectively during the time period between bucket midpoints T_k and T_{k+1} . The variable $CF_{i,c}$ denotes the net cash flows in each currency between time buckets k and $k + 1$.

To obtain the total ΔNII_t of all currencies on day t , all values have been converted to SEK, and then aggregated by

$$\Delta NII_t = \sum_{c=1}^3 \Delta NII_{c,t}.$$

In the same way as in the EVE calculations, the change in ΔNII over 90 days is calculated by subtracting the value of ΔNII on day 0 from the value of ΔNII on day 90.

Lastly, the $\text{VaR}_{0,99}$ is calculated in order to obtain the maximum loss not to be exceeded in 90 days

$$\text{VaR}_{0,99}(\Delta NII_{90}) = \min \{m : P(\Delta NII_{90} \leq m) \leq 0,01\},$$

where ΔNII_{90} denotes the 90 day rolling difference in ΔNII .

4.5 Backtesting

4.5.1 Application

Backtesting is a process that assesses the quality of a forecasting model by comparing the model's predicted values against actual values from a previous period. Although there is no regulatory guidance specifying the requirement for backtesting the CSRBB model, the model developed in this thesis is backtested to determine its predictive accuracy and if it is robust enough to handle changes in market conditions. The goal is to increase confidence in the model's ability to provide reliable estimates of risk exposure, which is critical for effective portfolio management. The idea is ultimately to compare the 90-days risk projection of the model with the factual observed 90-day changes of ΔEVE , and in this way estimate the forecasting ability of the model [17]. If the model consistently produces higher estimations compared to actual values, it would indicate that the model is overestimating the risk and could therefore be considered too conservative. Conversely, if the model consistently produces lower estimations, it indicates an underestimation of risk and can be said to be a too optimistic. For the purpose of backtesting the risk model of this thesis, the CSRBB model was backtested against observed data from one year back, i.e. the full year of 2022.

4.5.2 Statistical Assessment

When assessing the backtesting of a model, it is important to consider the occurrence of outliers, which are deviations from the expected results. The probability of an outlier occurring can be expressed as $p_0 = 1 - \alpha$, where α is the confidence level of the Value at Risk (VaR) calculation. In this section, we assume a confidence level of 99 percent (i.e., $\alpha = 0.01$), meaning that an outlier is expected in 1 percent of the observations [27].

We use the null hypothesis $H_0 : p \leq p_0$, where p is the observed fraction of outliers and p_0 is the probability of an outlier occurring. The null hypothesis is tested against the alternative hypothesis $H_0 : p > p_0$ to determine if the model accurately captures its risk estimation.

We compute the probability p of observing x outliers out of N observations using the binomial, or approximate normal, distribution. The binomial probability distribution

of observing exactly x outliers can be expressed as [30]:

$$p = P(X = x|N, p) = \binom{N}{x} \cdot p_0^x (1 - p_0)^{N-x}.$$

The probability of observing less than or equal to x outliers is given by:

$$P(X \leq x|N, p) = \sum_{X=0}^x \binom{N}{x} \cdot p_0^X (1 - p_0)^{N-X}.$$

The probability of observing more than x outliers is given by:

$$P(X > x|N, p) = \sum_{X=x+1}^N \binom{N}{x} \cdot p_0^X (1 - p_0)^{N-X}.$$

A variation on the binomial test proposed by the Basel Committee from the Basel II regulation [8] (Annex 10a on *Supervisory Framework for the Use of "Backtesting" in Conjunction with the Internal Models Approach to Market Risk Capital Requirements*) is a so-called "traffic light" or "three-zone" approach. It categorises the backtesting results into three levels: green, yellow, and red. These three zones can be defined in terms of the cumulative probability function of a binomial variable $F(x|N, p)$:

- **Green Zone:** $F(x|N, p) \leq 0.99$. This zone indicates an acceptable VaR model, with outliers ranging from zero to the beginning of the yellow zone.
- **Yellow Zone:** $0.99 < F(x|N, p) \leq 0.9999$. This zone indicates a risk model that may still be acceptable, as the violation count is not exceedingly high. More than x outliers are observed with a probability of 1%.
- **Red Zone:** $0.9999 < F(x|N, p)$. This zone indicates a model that may be rejected, as it is unlikely that too many outliers come from a correct VaR model. This zone begins at the point such that the probability of observing more than x outliers is 0.01%.

Having three levels as part of the backtesting evaluation allows for more subjective analysis of the model evaluation results rather than a simple accept or reject approach. In other words, one can avoid an erroneously rejection of the null hypothesis, known as a type 1 error.

Using a confidence level of 99 percent and a history of 2611 days (approximately ten years), one can calculate the following traffic light barriers:

$$\text{Green: } 0 \leq x \leq 28$$

$$\text{Yellow: } 29 \leq x \leq 38$$

$$\text{Red: } x \geq 39$$

Chapter 5

Results and Discussion

This chapter will present the results of the model applied on the studied balance sheet, as well as a discussion of choices made in constructing the CSRBB model. The projected cash flows and resulting yield curves are illustrated and commented on. Additionally, the model's validity is discussed by a comparison with FI's standardised model, as well as backtesting. Finally, some notes regarding the conservative aspects of the model is discussed.

5.1 Model Data

This section will illustrate and discuss the results of the projected cash flows, as well as the underlying yield curves of the credit spreads. These results are all fundamental to the results of the risk calculations, which will be presented in the subsequent section.

5.1.1 Projected Cash Flows

As the risk calculations of this thesis are highly sensitive to the balance sheet items, it is important to understand the balance between assets and liabilities in terms of the expected cash flows throughout the remaining lifetime of the instruments. Figure 5.1.1 illustrates the total expected cash flows from all balance sheet positions in both Swedish krona (SEK), Norwegian krone (NOK) and Euro (EUR). All currencies have been converted to SEK for a more comprehensive study.

As illustrated, there is an excessive number of expected cash inflows versus outflows.

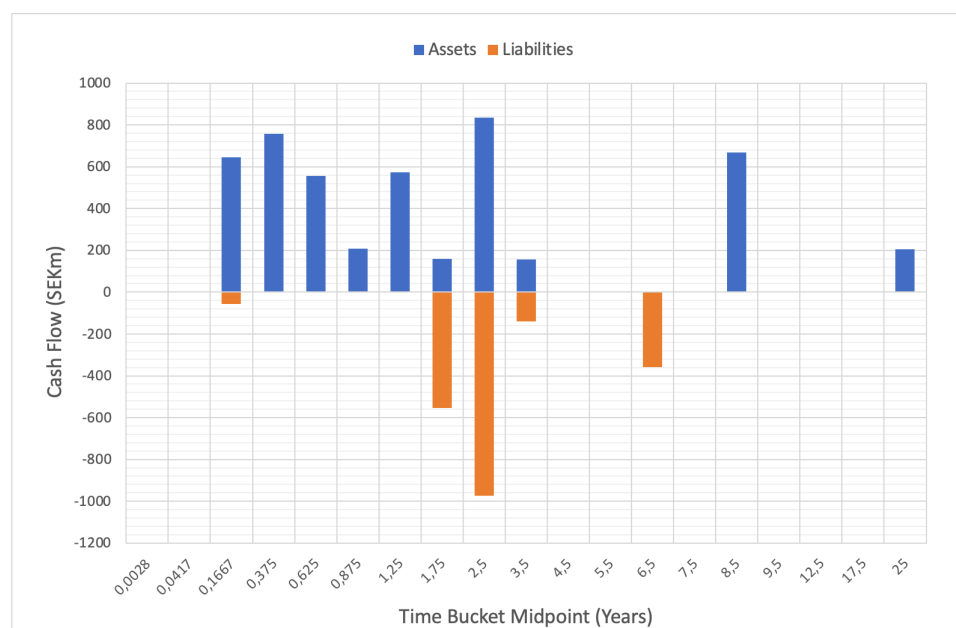


Figure 5.1.1: Total cash flows from all balance sheet positions.

Standard practice for banks is to have more cash inflows than cash outflows to maintain their financial stability and meet their financial obligations. It can also be seen in the figure that most positions of the balance sheet will mature within the next five years.

5.1.2 Yield Curves

Results of NSS

The yield curves plotted in the graphs below are based on daily sets of historical yield data spanning from 2013 to 2023. Since the NSS method performs particularly well for yield estimations beyond the data range, i.e. for longer maturities than what there is data for, yield curves can be extrapolated and used when yields for longer maturities are needed.

The number of data points available for each bond will determine the degree of under- or over-fitting the yield curves. In the case of Swedish mortgage bonds, figure 5.1.2, there were daily data for only two maturities (2 and 5 years) available, contributing to rather flattened curves.

Although the Nelson-Siegel-Svensson method is a well-suited option when extrapolating for longer maturities, it is considered weak in extrapolating for shorter maturities. Since only Swedish OIS data for maturities longer than one year

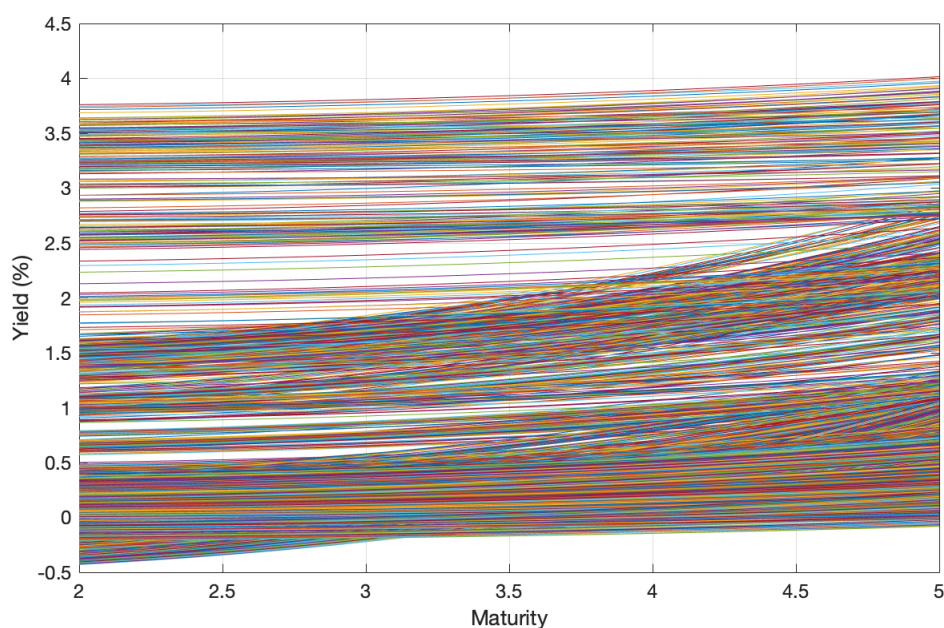


Figure 5.1.2: Historical daily yield curves of Swedish mortgage bonds.

were available, the Stibor interest rate with maturities as short as one week was added to the data set. The yield curves of the Swedish Overnight Index Swap (OIS) with the Stibor rate added are illustrated in figure 5.1.3.

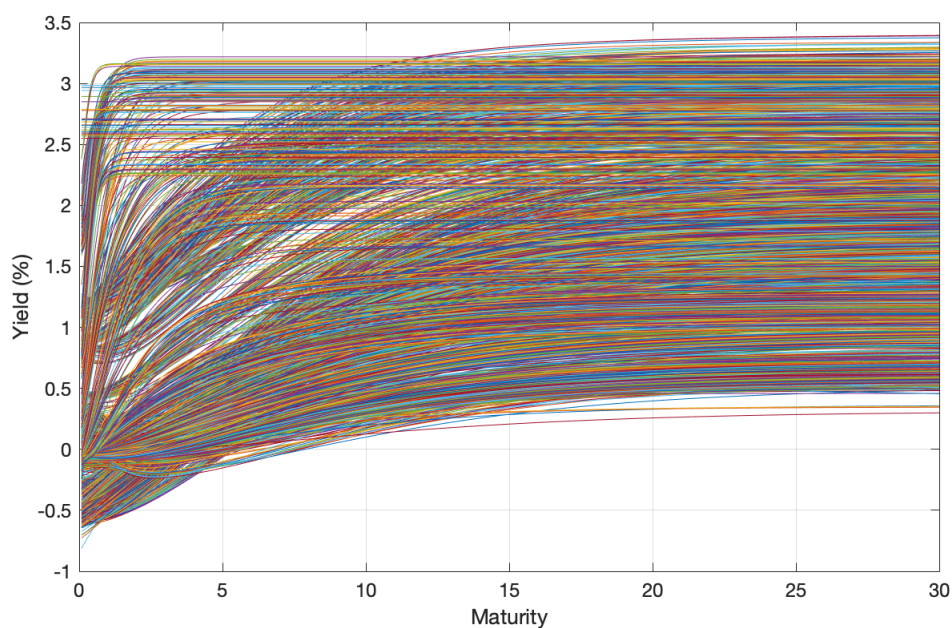


Figure 5.1.3: Historical daily yield curves of Swedish OIS and Stibor.

Choice of Risk-Free Curve

When constructing the risk model, there was a challenge encountered in the selection of the risk-free curve. There is no industry standard approach, thus it is up to the risk manager to choose the most suitable. The OIS and government bond yield curves are generally considered risk-free and accepted as two alternatives in the financial industry. EBA does not include any specifics on the choice of a risk-free benchmark in the regulations other than it should not include instrument-, sector- or entity specific credit spread or liquidity spreads.

FI includes government bonds as one risk category, which would argue in favor of the OIS curve being the best risk-free option. However, adding Stibor in the OIS curve construction introduces a major issue in modeling the credit spread risk of floating rate instruments with Stibor as the underlying instrument, especially in the NII calculations. In calculating the credit spread, the risky yield is reduced with the presumably risk-free, thus the credit spread would be zero for shorter maturities.

In order to mitigate this issue, the risk-free curve was constructed with government bonds. FI includes government bonds as a risk category in their model, however issuers with second highest and highest credit rating has a stress factor of zero, and therefore they are considered not to introduce any significant credit spread risk. A drawback of constructing the risk-free curve with government bonds is that some government issued zero coupon bonds are excluded from the CSRBB calculation since their resulting credit spread are zero. The best solution is still up for debate, however considering the high amount of floating rate instruments following Stibor in the studied portfolio, this was concluded to be the best solution.

5.2 Credit Spread Risk

5.2.1 EVE and NII

The results of the risk measures of the model can be observed in figures 5.2.1 and 5.2.2. The VaR for EVE and NII provide the potential loss in the value of the portfolio and profitability respectively. The resulting values of ES are consistently higher than those of VaR, which is to be expected. The important limitation of VaR is the risk of underestimating the capital needed in the case of an extreme event. For

example, two portfolios could have the same VaR, but one of the portfolios could be considered riskier if the expected loss in value or profitability is higher in case of the VaR cutoff point being crossed, i.e. the Expected Shortfall. ES captures the loss to be expected if one were to encounter an extreme event and is therefore more conservative. However, as shown in figures 5.2.1 and 5.2.2, the VaR and ES values are relatively close, indicating that historical changes in credit spreads over 90 days are relatively consistent. However, it can be seen that the VaR and ES for the EVE measure is greater than for NII, suggesting a greater loss in the value of the portfolio.

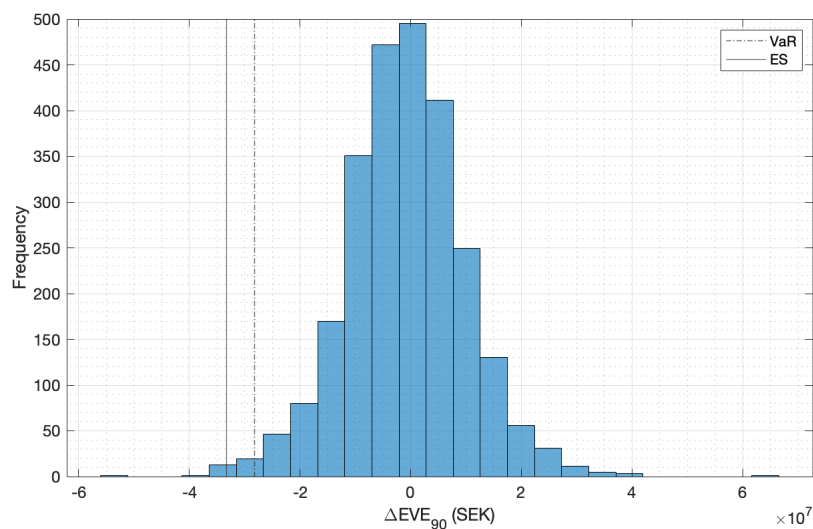
Instrument Type	VaR (SEKm)	ES (SEKm)
Total bonds	27.351	28.976
Total all instruments	28.226	33.242

Figure 5.2.1: Credit Spread Risk by EVE measure.

Instrument Type	VaR (SEKm)	ES (SEKm)
Total all instruments	15.928	18.742

Figure 5.2.2: Credit Spread Risk by NII measure.

In figures 5.2.3 and 5.2.4, all model 90-day changes of the ΔEVE and ΔNII are stacked in their respective histogram. The VaR and ES cutoff points are highlighted in both figures, which represent the results given in figures 5.2.1 and 5.2.2 above.

Figure 5.2.3: Histogram of ΔEVE results with cutoff points.

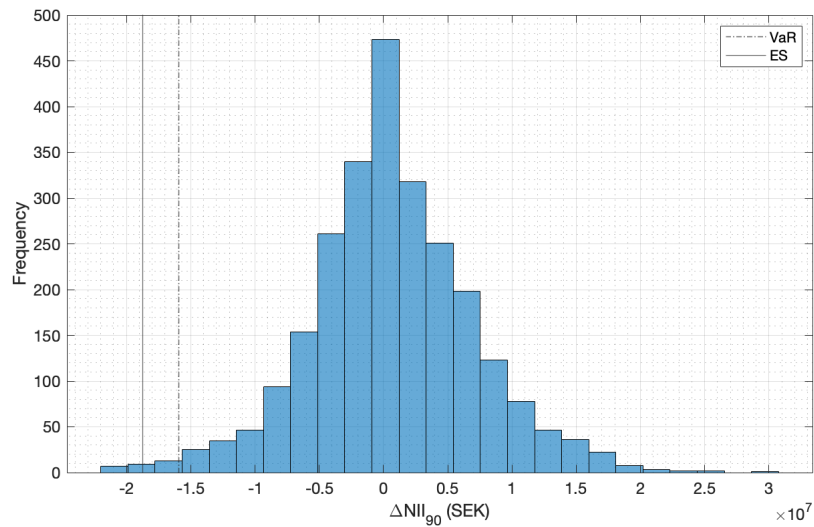


Figure 5.2.4: Histogram of ΔNII results with cutoff points.

5.2.2 Comparing Risk Measures

This subsection will discuss the choice between the two risk measures EVE and NII as the capital requirement.

An important question for risk managers is: which risk measure, EVE or NII, should be chosen as the capital requirement? In calculating the IRRBB, it is common in the industry to choose the largest of the two. As previously mentioned, the inclusion of liabilities in the CSRBB risk measures has not been as specified in the previous EBA guidelines. Thus, if choosing the largest value as the capital requirement, it is important to determine that the two measures provide a sufficiently complete risk profile by complementing each other. By complementing each other, it is meant that the two risk measures capture the risk induced by credit spread in all possible movements, i.e. both widening and tightening.

In order to evaluate the choice of risk measure as a basis of the capital requirement, it is necessary to investigate the drivers of the two risk measures. The EVE and NII measures respond differently to movements of the CS. The EVE measure, i.e. the loss in value of the portfolio, is sensitive to CS widening and will increase with a widening CS.

In the case of NII, the effect of CS movements on the risk measure is not as clear. Previously mentioned in subsection 4.4.2, the NII is calculated by subtracting income

expenses from interest income. In equation (4.3), one can observe that an increase in the forward rate will result in an increased size of both negative and positive cash flows. Thus, credit spread widening would result in both larger interest income and interest expenses. However, depending on the nature of the portfolio, the interest payments of the interest-bearing liabilities might not increase as much as the interest income of the assets and this would result in an increase of the net interest income between two time points. In contrast, there would be a decrease in the net interest income if the credit spreads were to tighten, resulting in the interest of the interest-bearing assets to decrease more than the interest of the interest-bearing liabilities.

It is clear that the outcome of the NII measure is highly dependent on the content of the portfolio. Impacting factors include the sizes and frequency of the interest income and payments as well as the dynamics of the underlying floating benchmark rate. The floating assets and liabilities on the studied balance sheet follow either Stibor 3M or Nibor 3M and are thus similar in dynamics and coupon frequency. Therefore, one could argue that the main contributing factors to the NII outcome are the amount of assets and liabilities as well as their coupon sizes. In fact, the studied balance sheet includes a much larger portion of floating assets in terms of nominal value compared to floating liabilities, therefore a widening in CS would result in a larger increase in interest income compared to interest expenses, thus an increase in NII between two points in time. This implies that a CS tightening will result in a decrease in NII and therefore a greater loss in profitability during 90 days.

It can be concluded that the two risk measures, NII and EVE, are driven by the credit spread movement in opposite directions. The loss in value of the portfolio will increase with a widening CS and the loss in profitability will increase with CS tightening. The risk of loss captured in the risk measures will thus not occur at the same time, therefore institutions should use the largest risk value of the two as the capital requirement. As observed in figures 5.2.1 and 5.2.2, the EVE provides the largest total VaR value of 27.4 SEKm compared to 15.9 SEKm in NII.

The strong dependence on the contents of the portfolio could be considered a weakness of the NII measure. The scenario of an institution having a portfolio with a larger share of floating liabilities than floating assets, will decrease in profitability when CS widens. In this scenario, both the EVE and NII measure is sensitive to CS widening, and choosing the larger measure of the two would thus not completely capture the

CSRBB.

5.3 Model Validity

To verify the model's predictive ability, the credit spread risk of the analysed portfolio was assessed by applying the FI standardised model and by performing backtesting.

5.3.1 FI's Standardised Model

The credit spread risks of the portfolio, as determined by the thesis model and FI's standardised model, are presented in figure 5.3.1. As the FI risk model is limited to the inclusion of assets only, the FI results can easily be compared to the Δ EVE results of the model of this thesis, as these also only include the assets of the balance sheet. Additionally, as the FI model is further limited to only include bonds, the thesis results used for the comparison to the FI result is the total Δ EVE for Swedish and Norwegian bonds. As figure 5.3.1 shows, the FI model resulted in a significantly larger credit spread risk compared to the thesis risk model, and is therefore a more conservative risk model. In industry, the FI model is considered to be a conservative model, particularly in terms of covered bonds.

Instrument type	CSRBB (SEKm)	
	VaR	
Constructed model (bonds)		27.351
	ES	28.976
FI model	38.353	

Figure 5.3.1: The CSRBB of the thesis model and FI's standardised model.

While the FI standard model is considered conservative, some financial institutions may prefer to develop their internal risk models, resulting in a lower risk capital margin while still ensuring sufficiently conservative results. Although developing an internal risk model may take longer time, institutions can make modeling decisions according to their specific needs and choices. All modeling choices must however be adequately justified in accordance with EBA guidelines.

5.3.2 Backtesting

As part of the evaluation process of the developed risk model, it is important to ensure that the results of the VaR align with its specified confidence level. For example, as the VaR is conducted for the 99th percentile, the model results are expected to be above this percentile in only 1 percent of the estimations. This means that, on average, only 2 to 3 estimates per year (out of 250 trading days) should exceed the actual values, corresponding to the 1 percent level.

Illustrated in figure 5.3.2 are the model predictions for the entire year of 2022, with VaR and ES levels shown in red and yellow, respectively. The blue graph in the same figure represents the actual change in ΔEVE over 90 days of the portfolio during the same period.

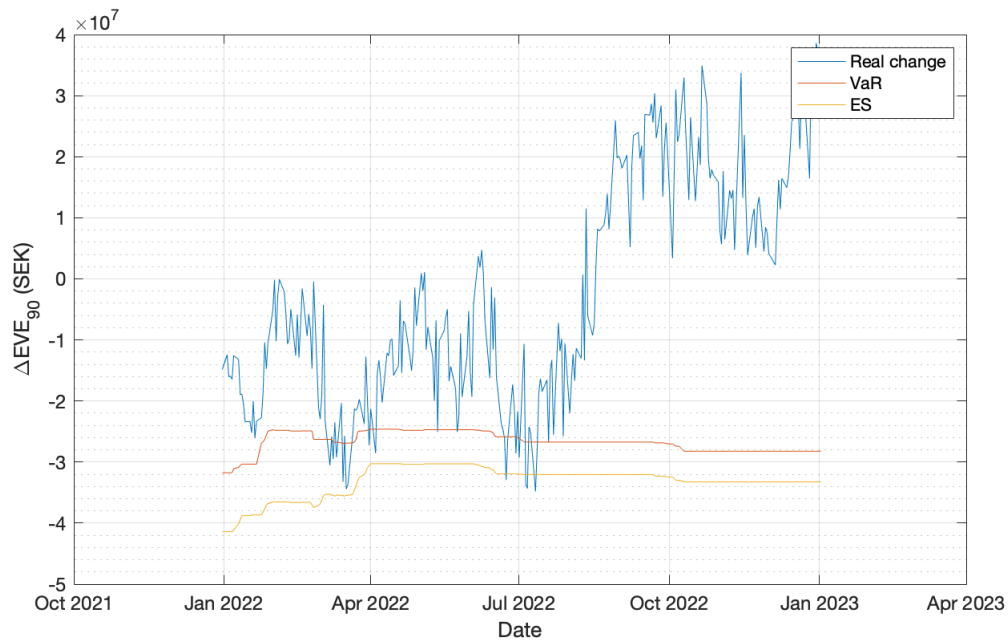


Figure 5.3.2: Backtesting model for EVE measure.

Displayed in figure 5.3.3 are the historical movement of ΔEVE over 90 days used in the backtesting. As the ten-year window of data progresses, the VaR and ES values are expected to change as any credit spread pikes appear or disappear from the data window. As illustrated in figure 5.3.2, the model increased its VaR and ES values between January and April. As seen in figure 5.3.3, this is probably due to the exclusion of large negative changes in ΔEVE values in the far left of the 10-year window of data. One conclusion is that as soon as any stressed historical time-points are introduced or no longer included in the data set, the model's predictions can change dramatically.

However, by choosing a sufficiently large time-interval, one can assume the occurrence of multiple financial stressful periods. Overall, the number of outliers during the model predictions over 2022 was counted to a total of 20, meaning the number of outliers is within the green zone of the traffic light assessment. This indicates that the model in fact can make predictions with 99 percent accuracy.

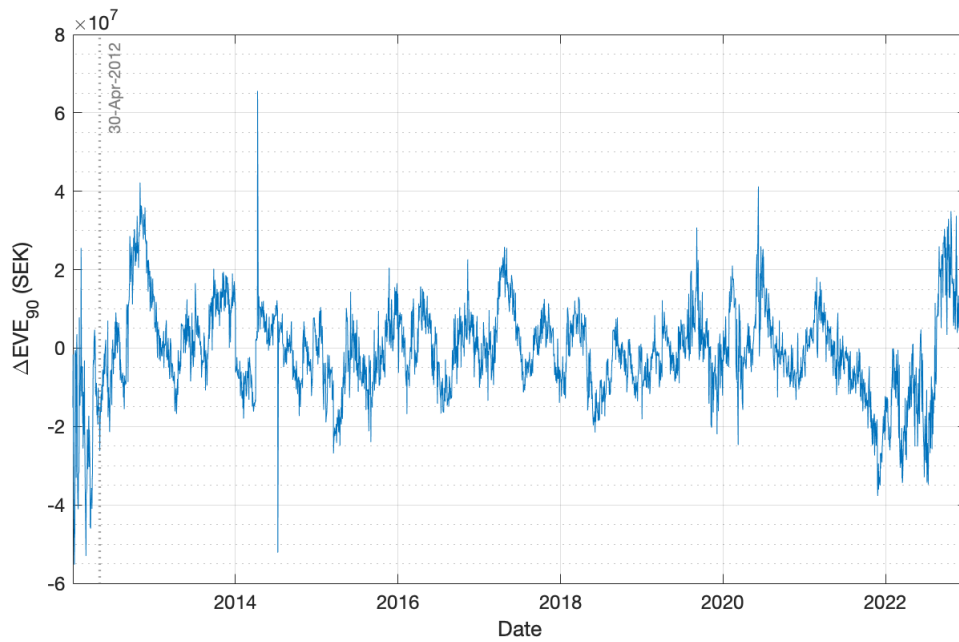


Figure 5.3.3: Historical ΔEVE change over 90 days.

The same analysis can be made for the backtesting of ΔNII_{90} . As seen in figure 5.3.4, the changes in model predictions in terms of VaR and ES depend on the historical movements in credit spread over 90 days in the historical 10-years window of data. For example, the increases in VaR and ES after October month are most likely due to the exclusion of large changes in credit spread observed in early 2012. This is a clear indication of how dependent the model is on the historical data included. If model data was only to be taken, say two years back, the model may undermine the possibility of sudden financial events occurring, which may greatly undermine the risk model predictions. However, it may also be argued that great financial stresses far back in time may influence current predictions too much. For example, if the entire year of 2012 would be excluded from the model, the risk from a NII perspective would be far smaller.

For the ΔNII_{90} backtesting, the number of outliers during the model predictions over 2022 was counted to a total of 0, which indicates that the number of outliers fall into the

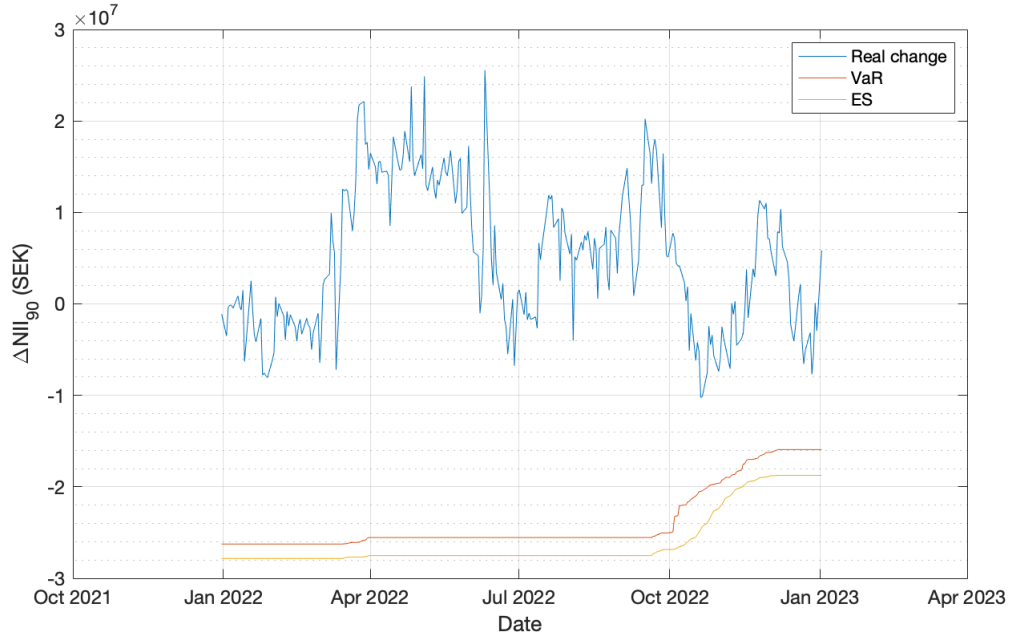
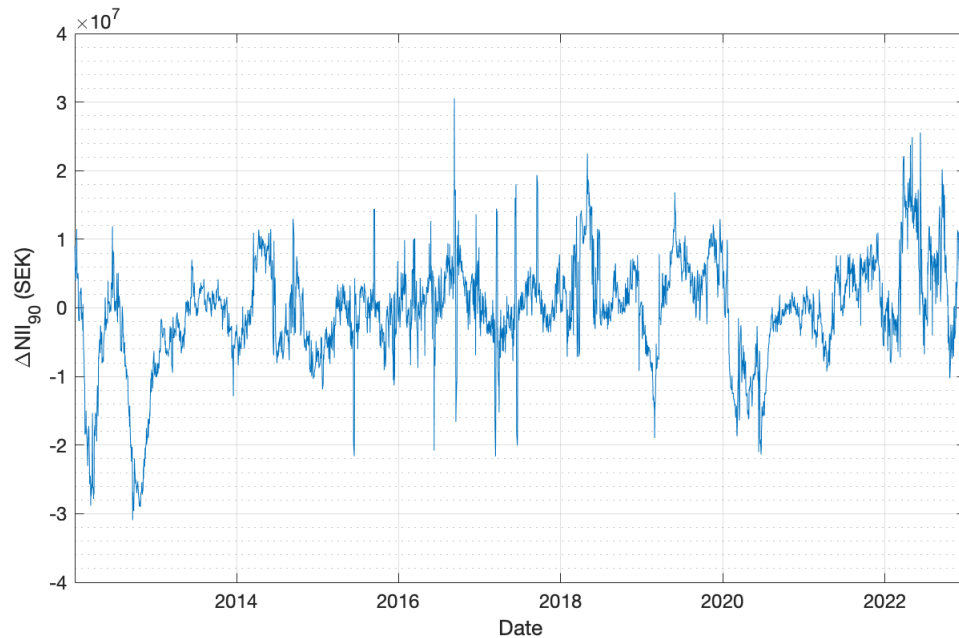


Figure 5.3.4: Backtesting model for NII measure.

green zone and thus the model predictions are deemed acceptable, as it indeed predicts with 99 percent accuracy.

Figure 5.3.5: Historical ΔNII over 90 days.

Finally, one can note that the final values of the VaR and ES in the EVE and NII backtesting graphs (i.e. final values of the red and yellow graphs) are the same values as the total VaR and ES results reported in figures 5.2.1 and 5.2.2.

5.4 Possible Model Adjustments

The following section will discuss possible adjustments to the model in order to obtain more conservative results.

In the construction of the model, several decisions were made affecting the results. It is possible to do adjustments to the model in order to be more conservative in the credit spread risk evaluation.

5.4.1 Risk Calculations

Firstly, a choice between two different approaches to calculate the VaR was made. The first approach is to calculate the VaR of the credit spread changes for each asset and liability instrument separately, from this the "worst" movement in credit spread for each instrument is captured. The second approach is to calculate the VaR on the portfolio risk measures, EVE and NII, thus obtaining the value and profitability of the portfolio not to be exceeded. Previously mentioned in subsection 4.3.1, widening or tightening of credit spreads are induced by extreme events affecting market conditions, such as the Great Financial crisis, the Eurozone debt crisis and the Covid-19 pandemic.

Using the first approach, we obtain a more conservative model since it allows for using credit spread movement data induced by different historical extreme events. The second approach calculates the risk of the complete portfolio, thus only allows for events that have happened simultaneously historically can happen again, simultaneously. For example, the credit spread of a bond may widen as a consequence of an extreme event, whereas the CS of another bond may widen even more as a consequence of another extreme event not related to the previously mentioned event. The first approach would use data from the two different historical events, since they were the most extreme. The second approach would only use data from one historical event for which resulted in the greatest effect on the whole portfolio.

While the first approach can be considered advantageous from a risk modeling point of view, since it introduces new scenarios, the results were considered to be unnecessarily high when compared to FI standardised risk model. One could also argue that the probability of the extreme events to happen at the same time is extremely low.

Therefore, the second approach of calculating VaR was chosen.

5.4.2 Mirroring

In addition to choosing how to calculate the VaR, it is possible to construct a more conservative model by introducing mirroring. By mirroring credit spread movements, it is assumed that any change in value of the credit spread can be both negative and positive. This is considered to be reasonable since, credit spreads are subject to both widening and tightening. By mirroring the credit spread movements, the number of data points is increased, and ensures that simulations would not be limited to historical events only. For comparative purposes, mirroring of all resulting credit spread movements has been applied and compared to the results of the non-mirrored credit spreads. It can be seen in figures 5.4.1 and 5.4.2 that mirroring results in a highly conservative risk model, particularly in terms of EVE, compared to the model excluding mirroring.

Instrument Type	VaR (SEKm)	ES (SEKm)
Total bonds	26.093	27.842
Total all instruments	29.411	35.470

Figure 5.4.1: Credit Spread Risk for EVE measure with mirroring.

Instrument Type	VaR (SEKm)	ES (SEKm)
Total all instruments	16.630	19.440

Figure 5.4.2: Credit Spread Risk for NII measure with mirroring.

On a last note, due to the risk model being a historical VaR model, the model results are highly dependent on the historical time period chosen. This has also been discussed in subsection 5.3.2. Thus, it is possible to obtain a more conservative model by using bond data obtained for a longer time period than 10 years. By extending the time period from which historical data is obtained, more extreme events of economic stress will be included.

Illustrated in figures 5.4.3 and 5.4.3 are the 90-day changes in ΔEVE and ΔNII with and without mirroring. The VaR and ES cutoff points are marked for both the non-mirrored data (red) and the mirrored data (blue). It can clearly be seen that both risk measures increase in value when applying mirroring to the data.

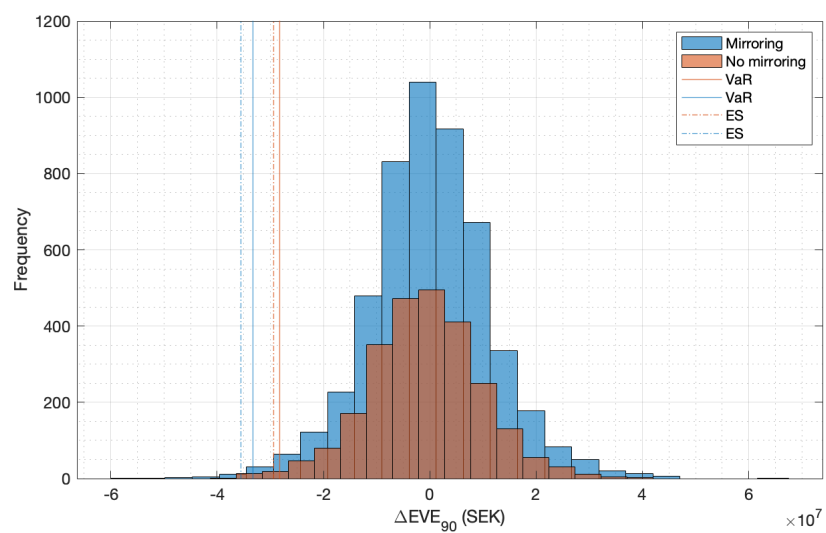


Figure 5.4.3: Change in ΔEVE over 90 days with and without mirroring.

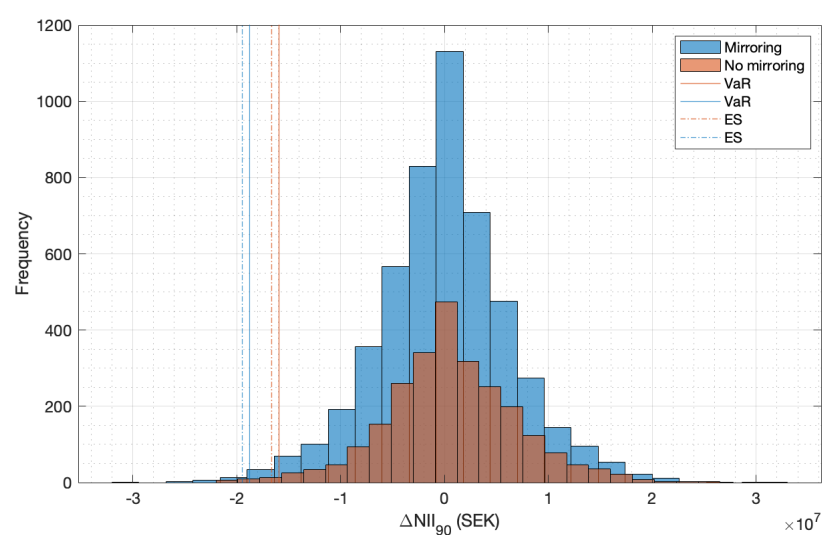


Figure 5.4.4: Change in ΔNII over 90 days with and without mirroring.

Chapter 6

Conclusions

The conclusions chapter of this thesis provides a summary of the key findings and insights of the research. The chapter also discusses the limitations of the study and potential future research.

6.1 Model Conclusions

This thesis has provided a model to calculate a financial institution's exposure to CSRBB. It will continuously take into account ten years of data as time progresses, and make a prediction for a given portfolio's CSRBB 90 days ahead of time. The measured risk is given in terms of VaR and ES, which both provide the resulting capital requirement in line with EBA regulations. The model allows for risk managers to make some choices of their own which may allow for more or less conservative results. For example, either the VaR or ES measures could be preferred and chosen as capital requirement depending on risk aversion.

As accentuated throughout the thesis, the model results are highly dependent on the balance sheet. Thus, the main objective has been the development and application of the model, in accordance with EBA directives, rather than the numerical results. The resulting VaR and ES values are, however, imperative for the final model evaluation and validation steps. By comparing the model results with the FI standardised CSRBB model, as well as backtesting results, the validity of the constructed CSRBB model was supported. However, it is advised to perform regular validation procedures of the model to ensure consistency.

The EBA guidelines are ambiguous and there is a lot of room for interpretation. Due to the lack of clear definitions and precise regulatory guidelines, several decisions were made based on the content of the studied portfolio. This means that the model may need adjustments for improvement if applied to other portfolios. For example, the choice between OIS and government bonds as the risk-free curve may be exchanged in the case of a portfolio with less floating rate instruments following the Stibor rate.

Additionally, a key finding during the project was the exclusion of liability instruments in the calculation of EVE. However, this is also dependent on the nature of the portfolio, as mentioned earlier in the case of re-emission of bonds or at the call date of the perpetual bond, some credit spread risk may be induced. Also, it was concluded that the choice of risk measure is also dependent on the nature of the portfolio. Since the portfolio had a large share of floating interest rate assets, the EVE and NII risk measures are sensitive to CS movements in the opposite directions which motivates using the largest of the two as the capital requirement.

The use of historical yield data in the model carries inherent risks, including the possibility of errors in the data and other risks in the manipulations made in the data processing. However, these risks were carefully considered when analysing the reliability of the model results. While there always is a trade-off between accuracy and precision in financial modeling, the model results were deemed accurate and precise enough to meet the objectives of this thesis.

One could wish for more specifics from EBA on the selection process of the credit spread sensitive instruments, particularly in terms of liability instruments, as well as how to handle liabilities in the calculations of the risk measures. It is possible that the short deadline for implementation of the guidelines (December 31 2023), in combination with ambiguous guidelines on CSRBB, will result in significant challenges for both banks and supervisory authorities.

6.2 Future Work

In this section some topics that are not addressed in this paper and are subject to further research will be presented.

- 1) In this model, the risk measures in each currency is converted to SEK. However, further research could be done on how to properly aggregate risk measures in

different currencies, and whether there is a correlation between FX risk and CS risk;

- 2) In modelling the CS risk, the choice of the risk-free benchmark curve plays a central role and has great impact on the end-results. Therefore, additional research on the most suitable risk-free curve, and a comparative study of different choices would be an interesting topic for further research;
- 3) Previously mentioned in subsection 3.3.3, Monte Carlo VaR methods are advantageous as they, in contrast to historical VaR, simulate new scenarios. It would be interesting to do a comparative study of the two VaR methods to see if a, for example, *GARCH* based Monte Carlo model would produce significantly different risk results compared to the model of this paper;
- 4) The model of this thesis is applied on one balance sheet only, and it would therefore be interesting to apply it on other balance sheets as well to see how the CSRBB is affected. Additionally, it would be interesting to see how CSRBB is affected when the holding periods of the bonds are adjusted.

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Appendix A

First Appendix

A.1 Model Data

A specification of the yield data used can be found in figure A.1.1 below.

FX	Instrument	Specification	Type	Special Treatment / Assumption	History
SEK	Floating Rate Instruments	Stibor Overnight Index Swaps 3M + Stibor	Yield	Stibor included in the Swap 3M dataset for tenors shorter than 1 year	01.01.2013
	Covered Bonds	Swedish Central Bank Mortgage Bonds	Yield		01.01.2013
	Region and Municipality Bonds	Swedish Regional Bonds	Yield to Maturity	A fix spread between regional and government bonds are applied before 30.09.2016	30.09.2016 (01.01.2013)
	Government Bonds	Swedish Government Bonds	Yield		01.01.2013
EUR	Floating Rate Instruments	Euribor Interest Rate Swaps 3M + Euribor	Yield	Euribor included in the Swap 3M dataset for tenors shorter than 1 year	01.01.2013
	Government Bonds	EuroZone Government Bonds	Yield		01.01.2013
NOK	Floating Rate Instruments	Nibor (Norwegian Interbank Offered Rate)	Yield		01.01.2013
	Covered Bonds	Norwegian Mortgage Bonds	Yield		01.01.2013
	Government Bonds	Norwegian Government Bonds	Yield		01.01.2013

Figure A.1.1: Yield data for all portfolio categories and currencies.

Since there is no single generic bond that represents Swedish regional bonds, a number of different bonds issued by Swedish regions and municipalities have been used. These bonds vary in terms of their maturity, issue date and expiration, meaning both active and expired bonds have been collected to obtain a long historic database of daily yields to maturity. As a second step, the yield to maturity for these bonds have been converted to yield rates for fixed maturities and approximated by taking the mean yield value between all papers available. The data is then used in the NSS function, introduced in subsection 3.2.1, to estimate the yield curves of the bonds. Again, this approach was necessary since the only data available for these papers were yield to maturity and not daily yield rates.

For historically unavailable data, i.e. data unavailable for a history of ten years, a methodology of using a “fixed” yield is applied. The fixed yield will be calculated in such a way that it is added to the yield of another risky bond for the dates with missing data. In this thesis, Swedish regional bonds are the instruments with missing data. A detailed description of how the fixed yield was determined is found in section A.2.

A.2 Fixed Yield Methodology

As previously mentioned in subsection 4.3.1, a methodology for applying a “fixed” yield for dates when historical data is missing has been applied to Swedish regional bonds. The Swedish regional bonds with missing yield data will be referred to as Bond 1. The steps to determine the fixed yield for Bond 1 are described as followed:

1. Choose another risky bond for which there are no historical data missing. For example, say this bond has ten years of daily historical data available. This bond will be referred to as Bond 2.
2. Collect the daily yield data for both Bond 1 and Bond 2. Say Bond 1 only has yield data for the past five years, and Bond 2 for the past ten years. Bond 1 is thus missing the first five years of yield data.
3. Compare the daily yields of Bond 1 and Bond 2 for the past five years. This is to say, when there is data available for both bonds. Calculate the daily difference in yield between the two bonds. Take the mean of the differences in yield between the two bonds. This mean represents the on-average daily difference in yield between the Bond 1 and Bond 2, called the “fixed” historical yield.

4. The fixed yield will then be applied to the first five years of data of Bond 2. Adding the fixed yield means one can approximate the yield of Bond 1 which is missing data during this time. This is to say, the fixed yield is added on to the first five years of the ten-year data set of Bond 2. These values thus effectively replace the missing historical data of Bond 1 and results in a full set of ten-year historical data.

In summary, this approach will have historical yields be represented by the yield movements of a proxy bond for which there is historical data available. Note that the calculated yields for missing dates will be composed of two parts, where one part is fixed and the other variable.

